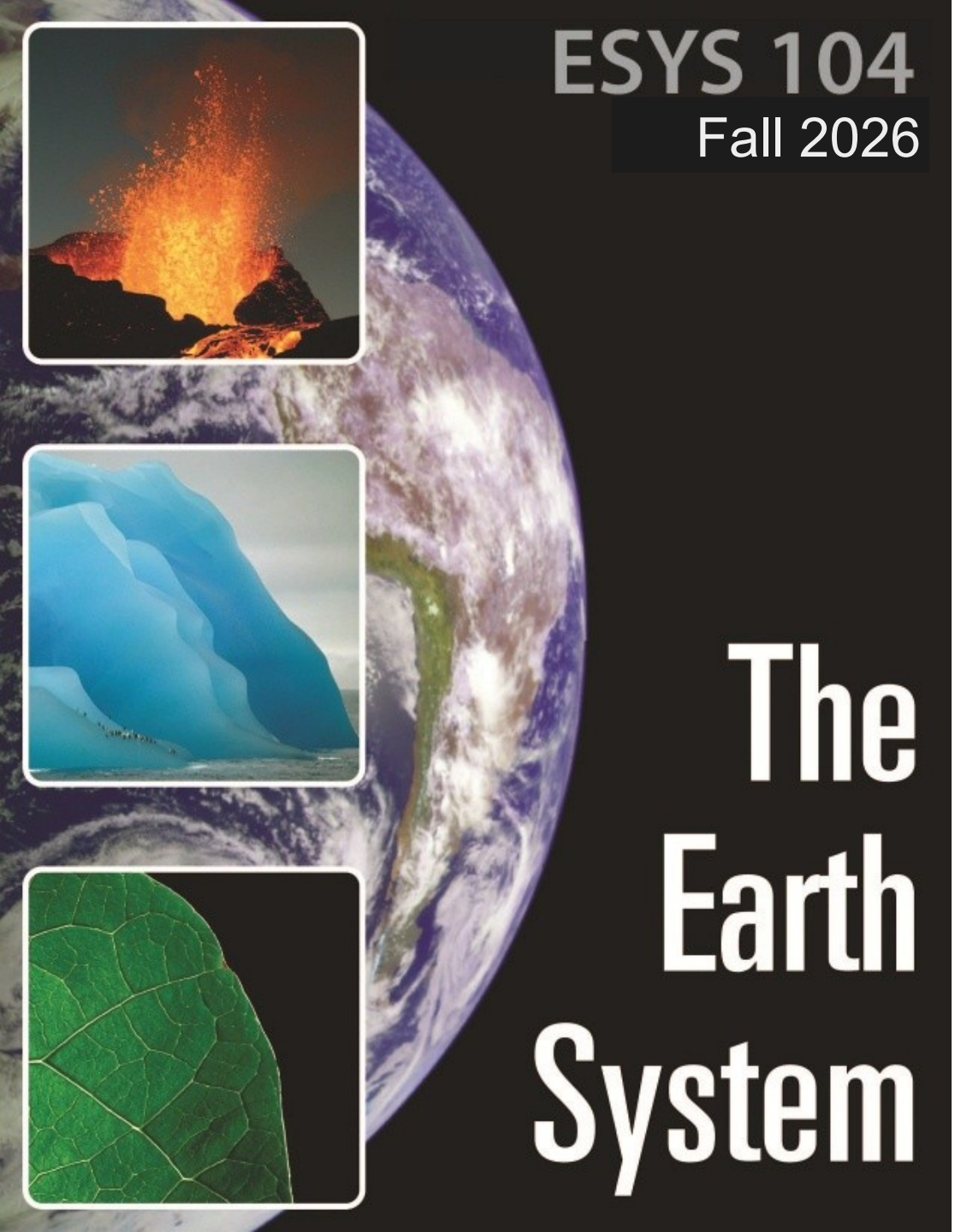
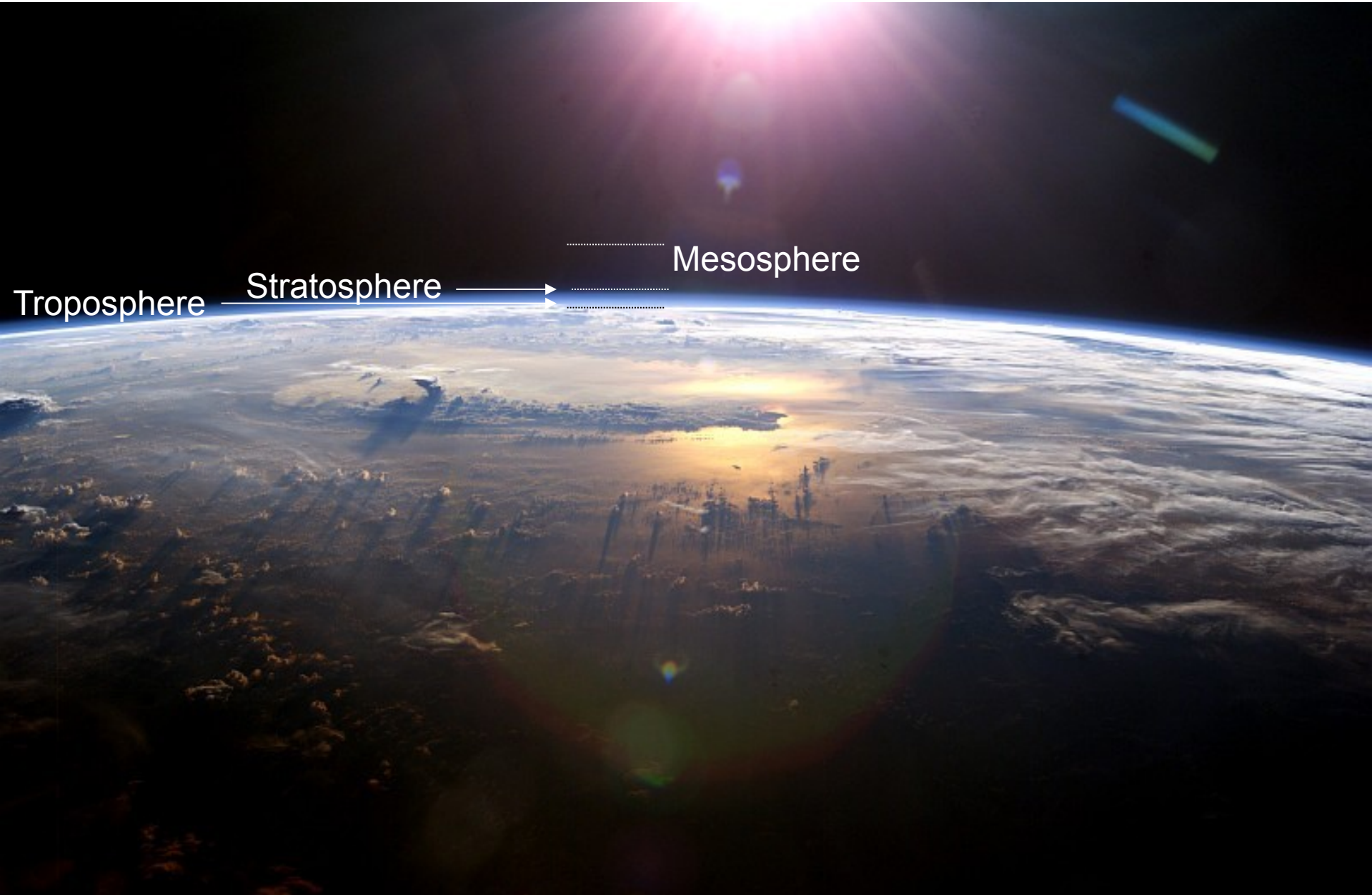


#6: Greenhouse gasses, Atmosphere circulation

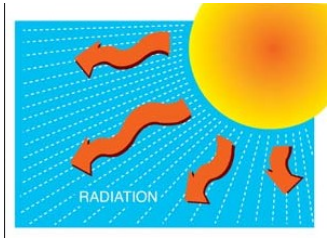
The Atmosphere

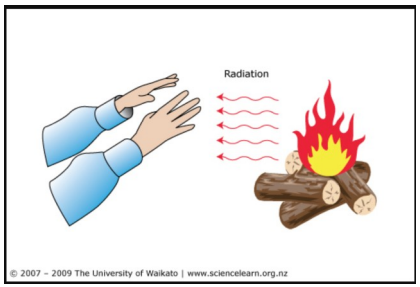


Energy Budget of the Earth

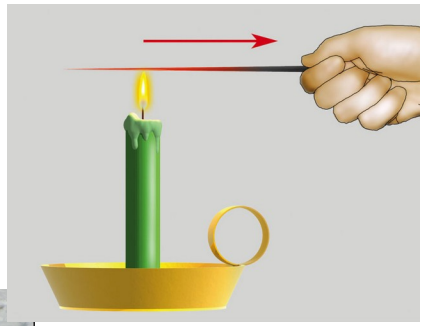
Sensible Heat

Radiation





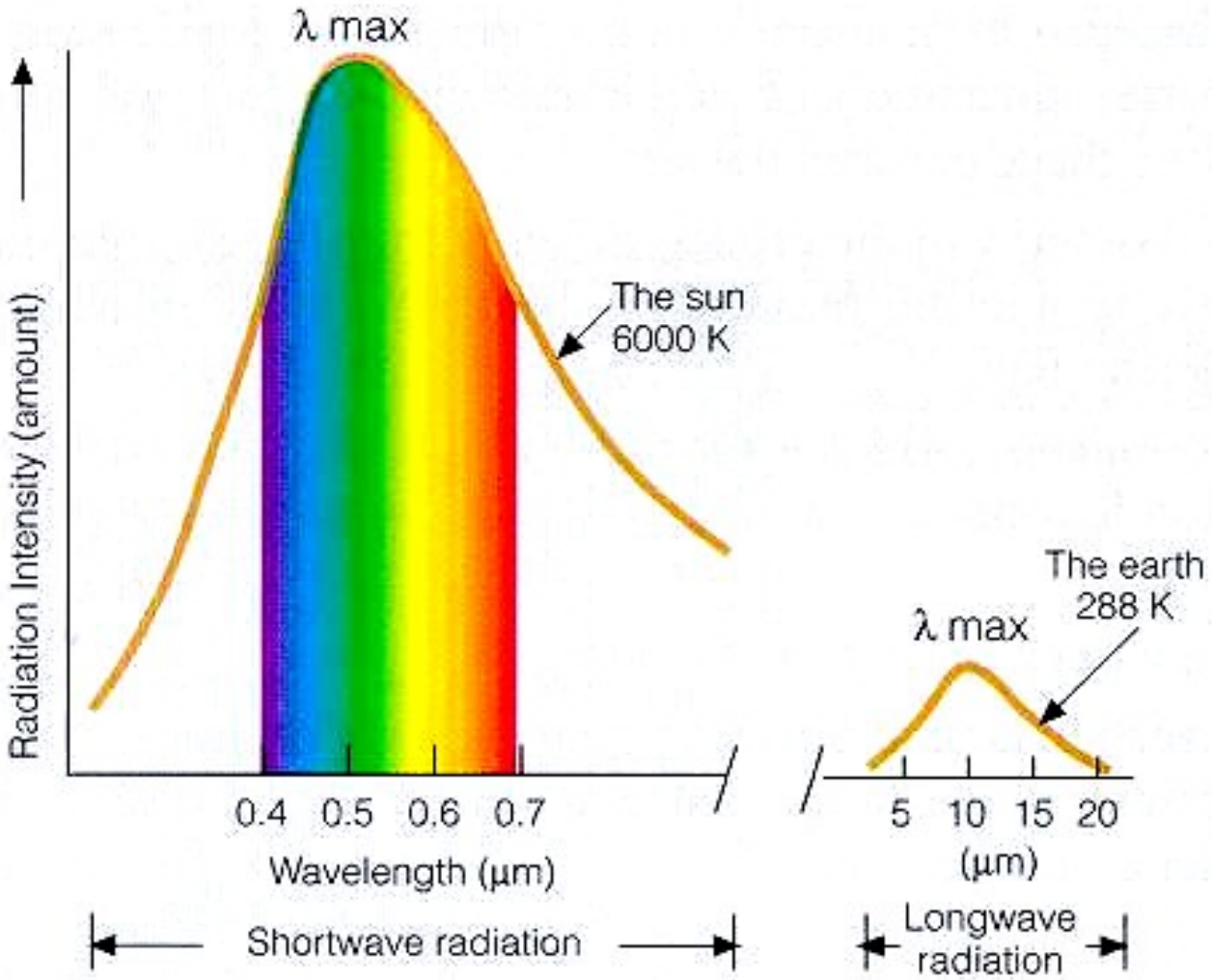
Convection



Conduction

Latent Heat

Radiation Properties: Sun vs. Earth



$T_{sun} \approx 5780$ K

→ Sun's peak emission is green

$T_{earth} \approx 288$ K

→ Earth's peak emission is in the infrared

Radiative Balance Terms

Solar constant at Earth's orbit S_0 = 1368 W/m^2

Albedo (reflectivity)

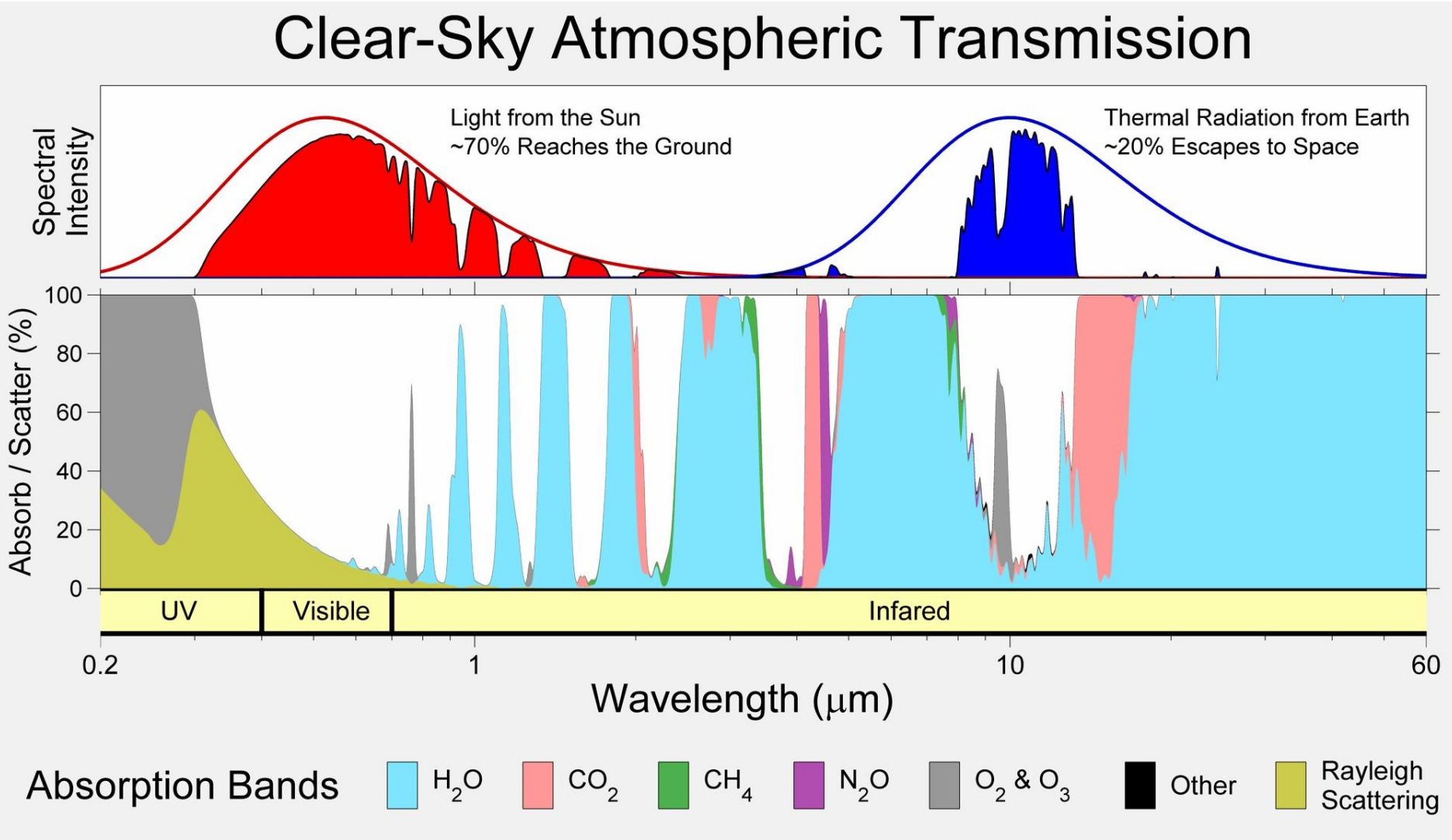
Atmospheric absorption
- short wave: ozone, particulates
- long wave: H_2O, CO_2, CH_4, etc.

Energy emitted: blackbody $E_{out} = \sigma T_E^4$

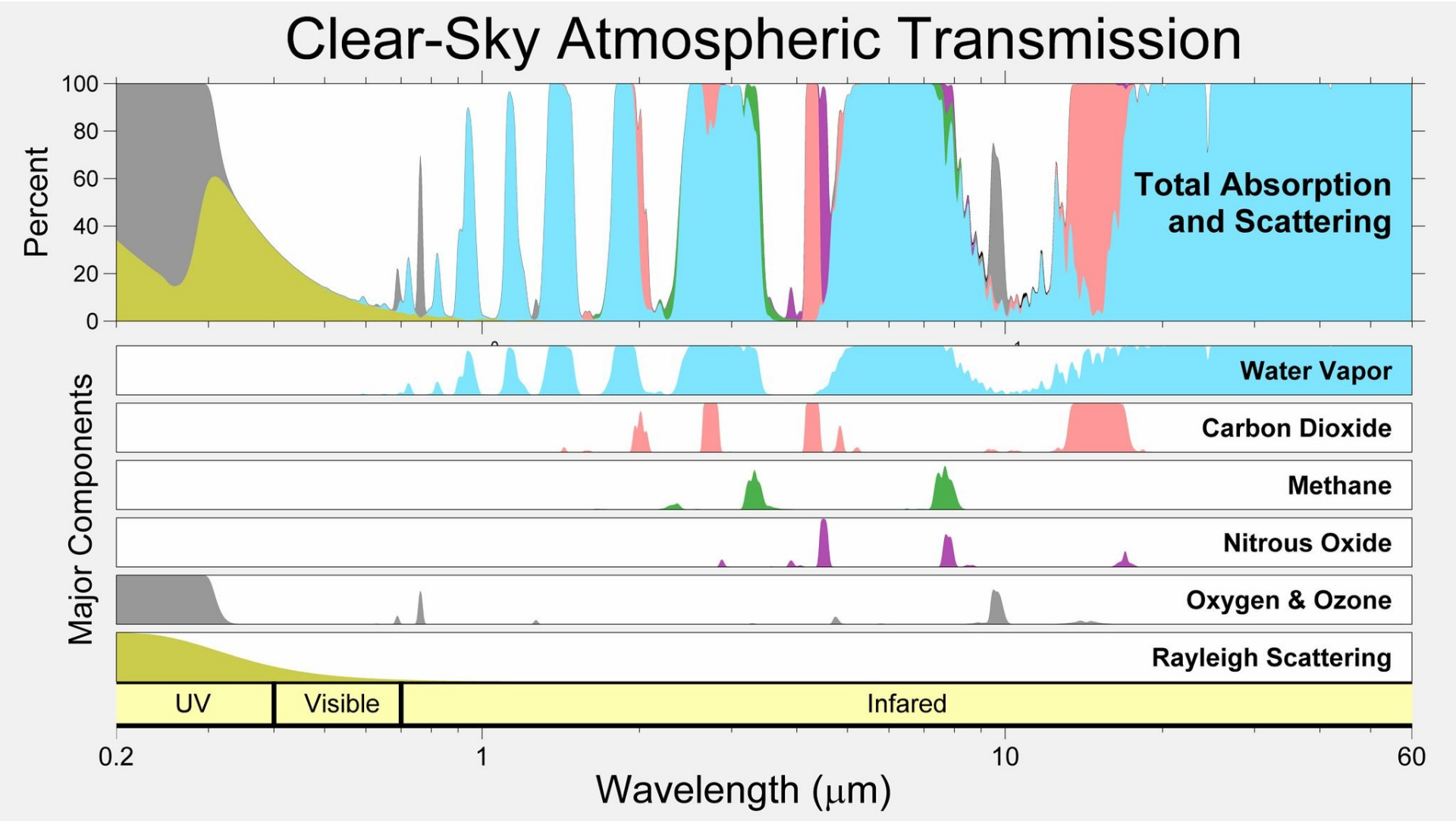
graybody $E_{out} = \epsilon \sigma T_E^4$

Stefan-Boltzman constant sigma σ = 5.67 x 10^{-8} W/(m^{2}K^4)

Emissivity (eps) ϵ from 0 to 1 (one is a perfect black body)

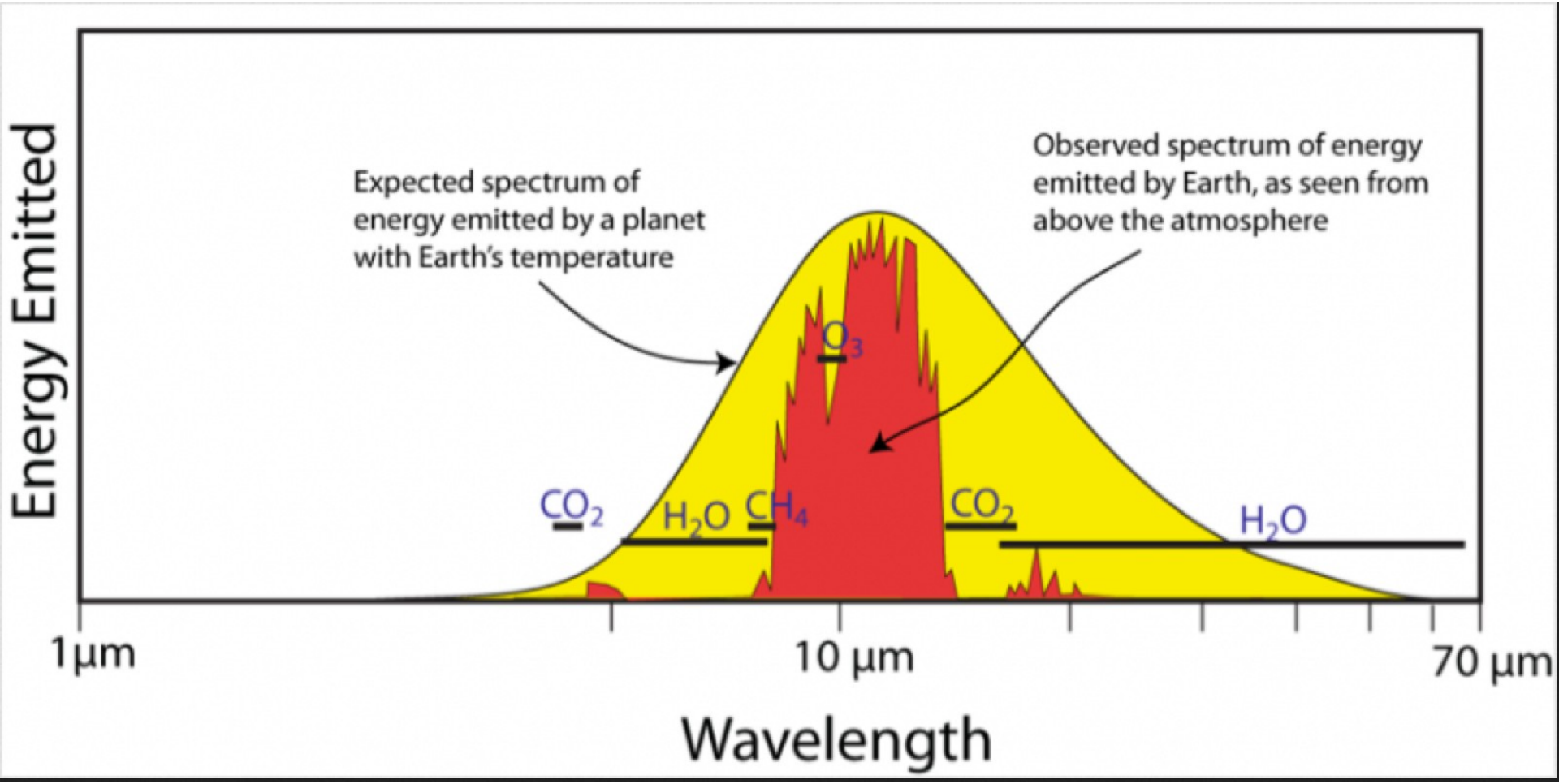

Clear-Sky Atmospheric Transmission
Spectral Intensity
Light from the Sun
~70% Reaches the Ground
Thermal Radiation from Earth
~20% Escapes to Space
Absorb / Scatter (%)
100
80
60
40
20
0
UV
Visible
Infared
0.2
1
10
60
Wavelength (μm)
Absorption Bands
H2O
CO2
CH4
N2O
O2 & O3
Other
Rayleigh Scattering

R A Rohde


Clear-Sky Atmospheric Transmission
Percent
100
80
60
40
20
0
Total Absorption and Scattering
Major Components
Water Vapor
Carbon Dioxide
Methane
Nitrous Oxide
Oxygen & Ozone
Rayleigh Scattering
UV
Visible
Infared
0.2
1
10
60
Wavelength (μm)

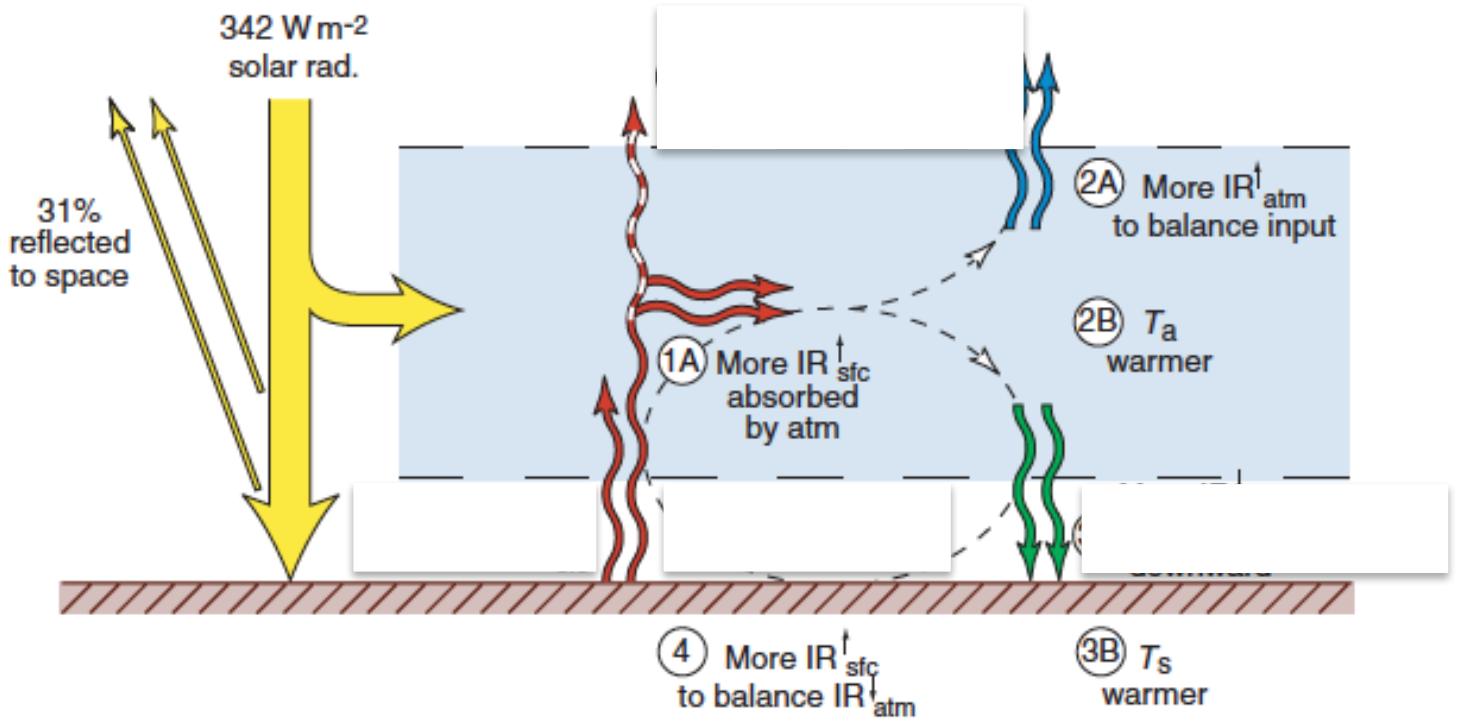
R A Rohde

Spectrum of Earth Radiation
Top of the Atmosphere

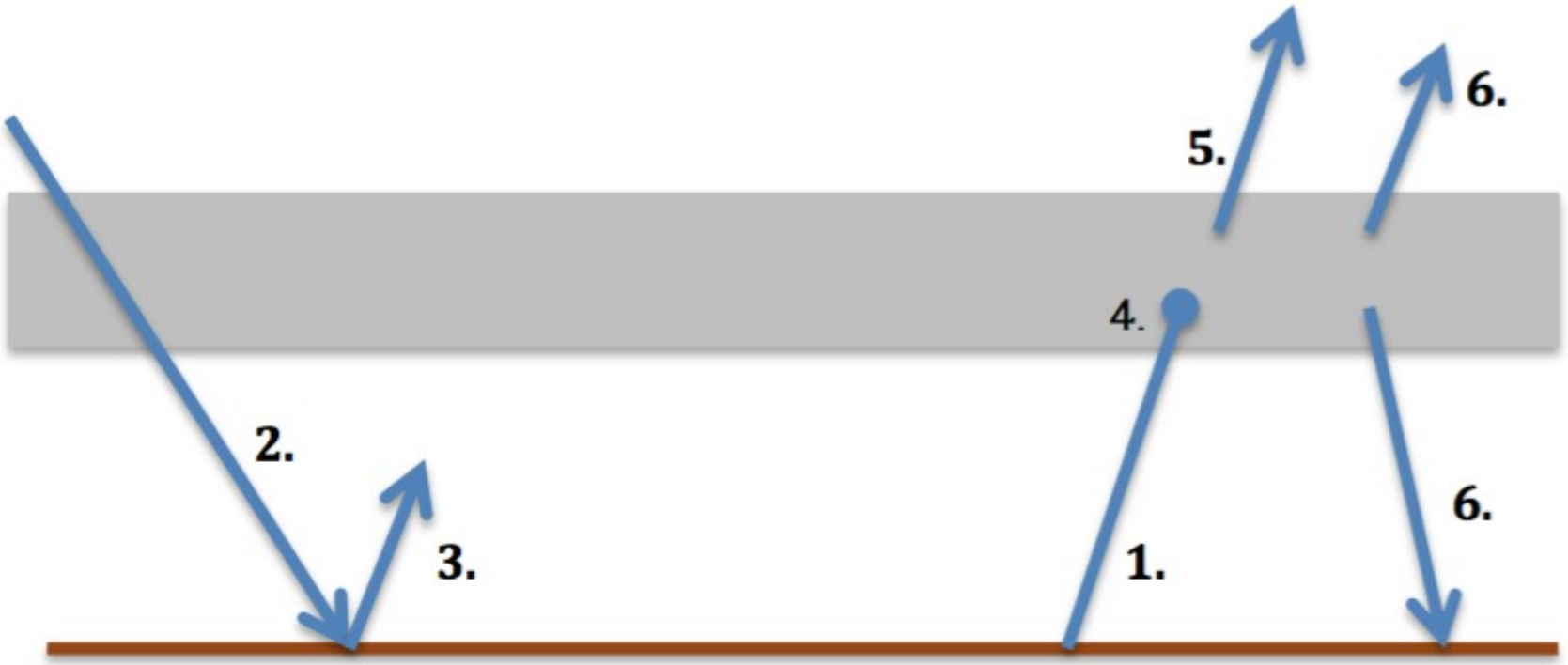


Energy Balance

Real Atmosphere – greenhouse effect



Atmosphere Energy Balance



Short wave (visible)

Long wave (thermal IR)

Atmosphere Energy Balance

Incoming solar shortwave radiation: $S_0/4$

Solar energy reflected back into space: the albedo α

Outgoing surface energy: σT_e^4

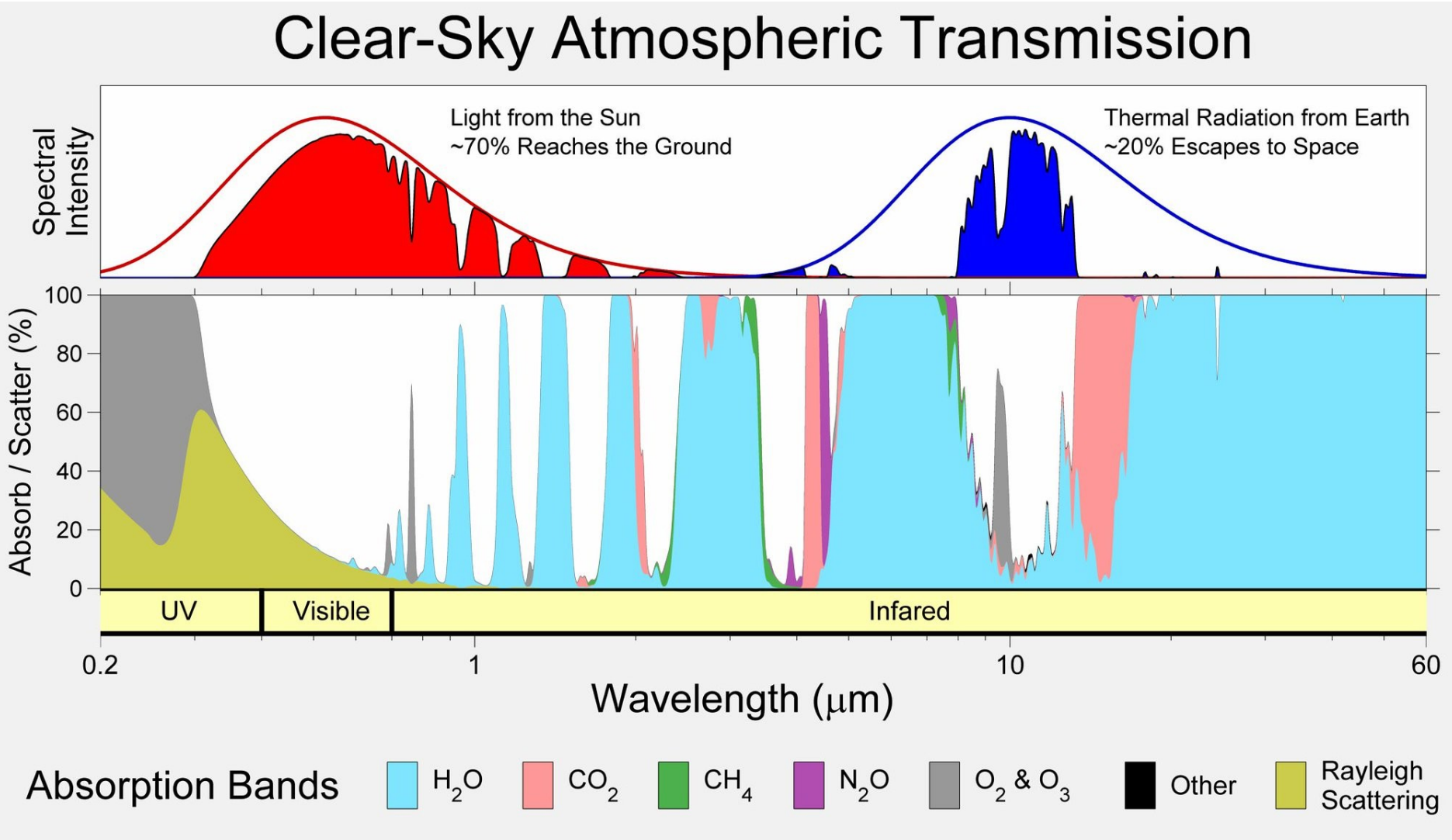
Surface longwave energy absorbed by the atmosphere: $\epsilon_a \sigma T_e^4$

Rest of the surface longwave energy not absorbed by the atmosphere: $(1-\epsilon_a)\sigma T_e^4$

Atmospheric longwave energy emitted: $\epsilon_a \sigma T_a^4$

Earth's surface energy balance:

$$S_0/4(1-\alpha)+\epsilon_a \sigma T_a^4=\sigma T_e^4$$


Clear-Sky Atmospheric Transmission
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100
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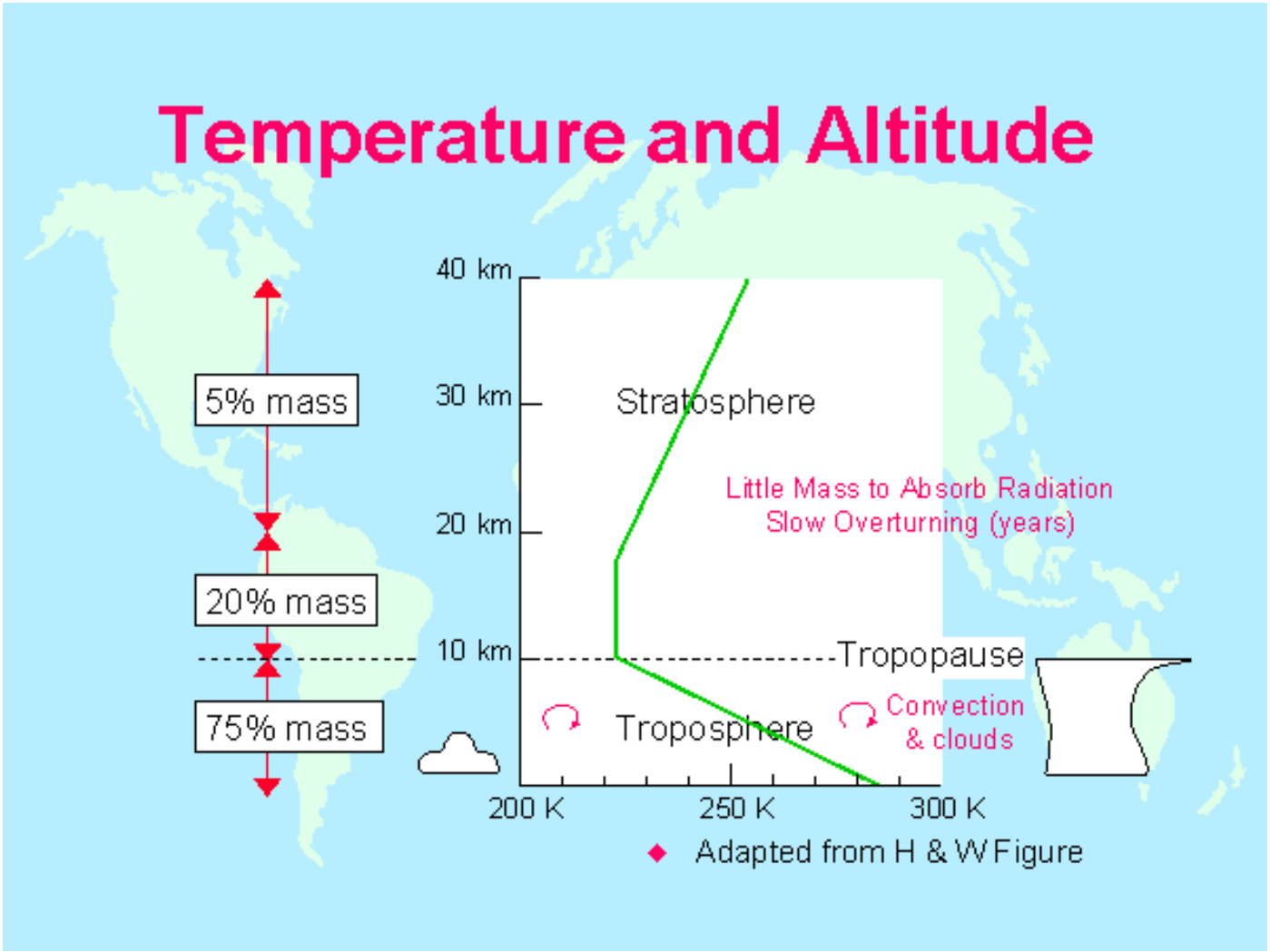
0.2
1
10
60
Wavelength (μm)
Absorption Bands
H2O
CO2
CH4
N2O
O2 & O3
Other
Rayleigh Scattering

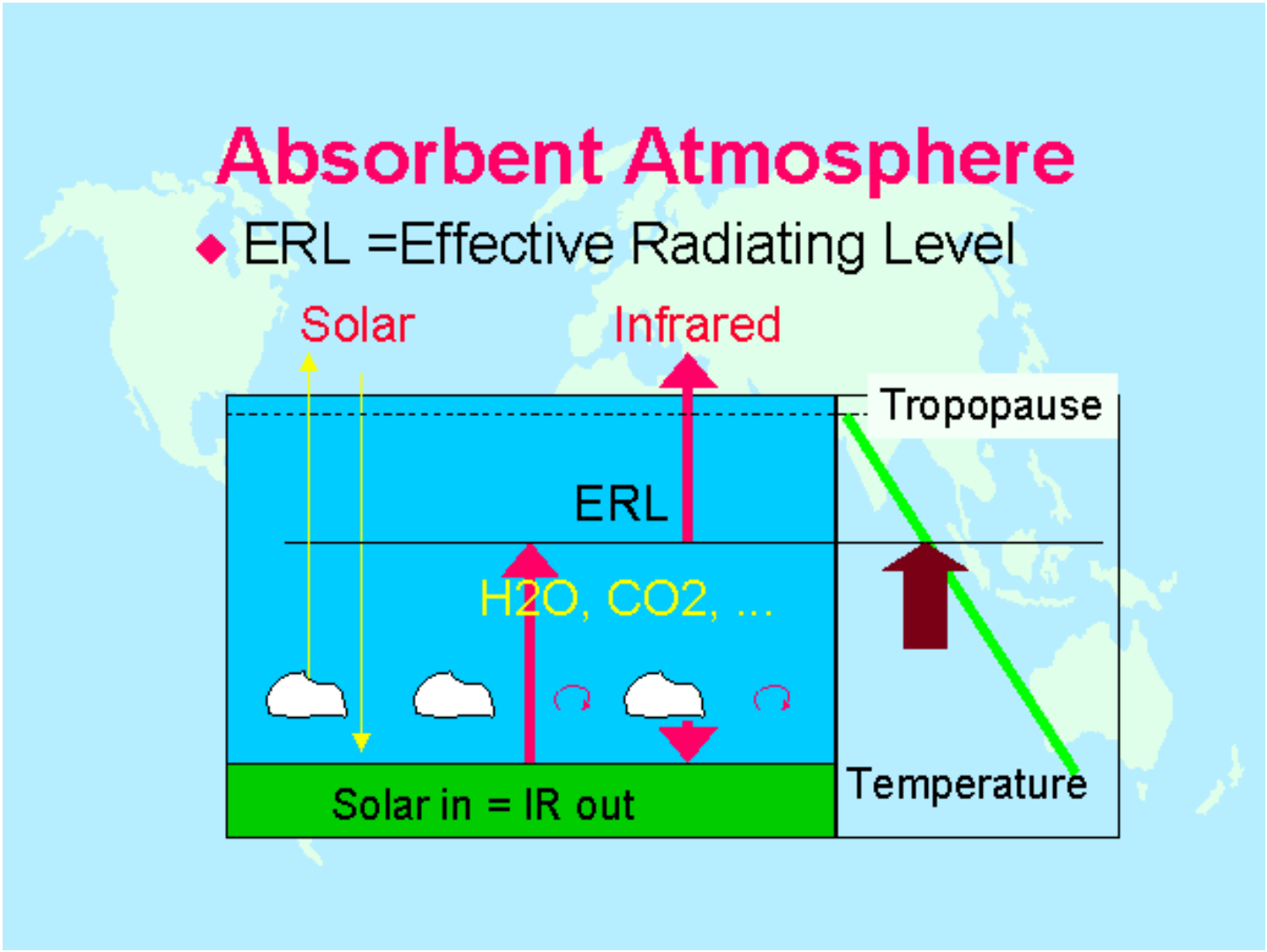
R A Rohde

Effective Radiating Level



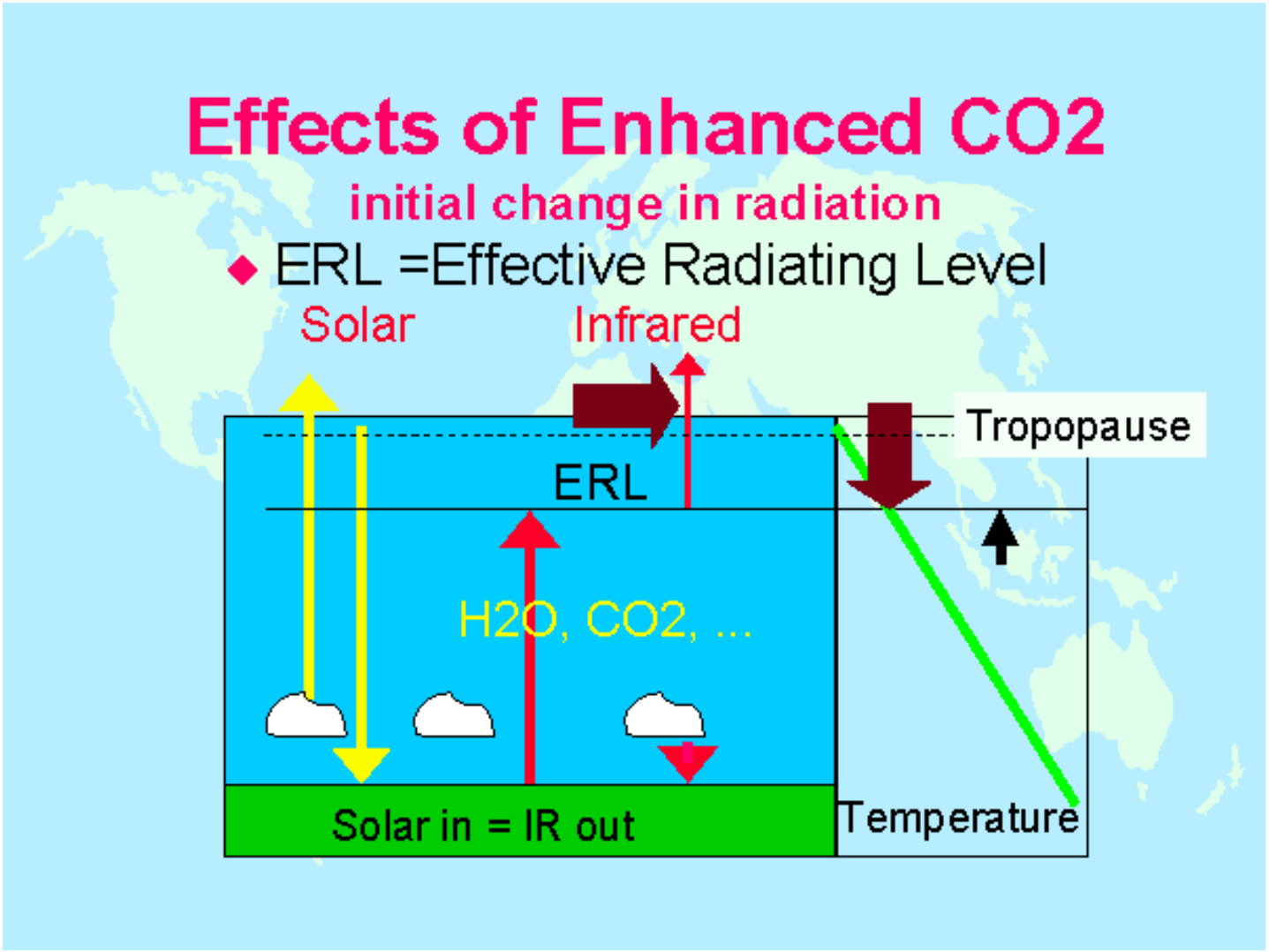
Generally around 8-10 km the atmosphere becomes mostly transparent to long-wave radiation

Effective Radiating Level



ERL: Generally around 8-10 km

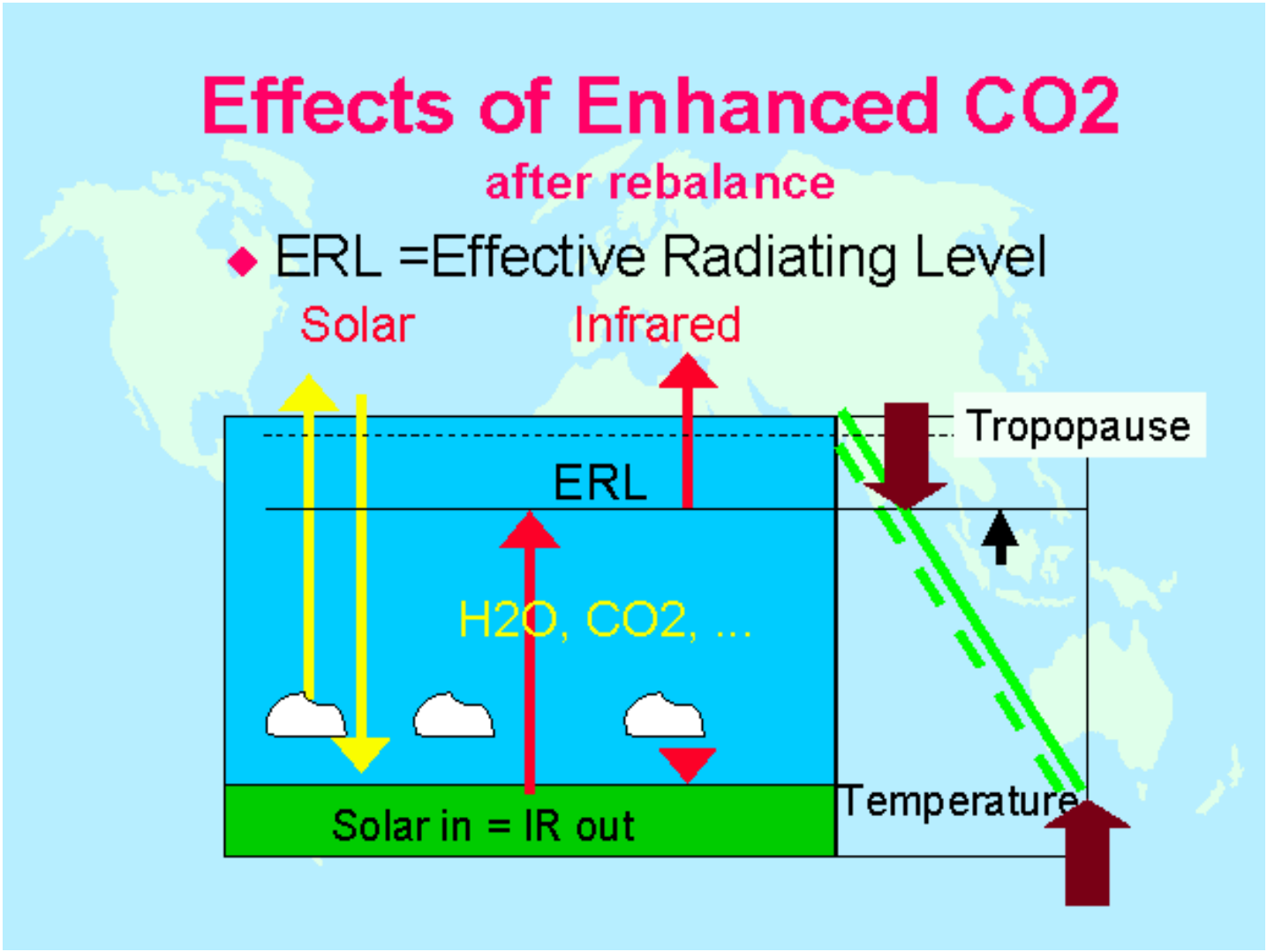
Effective Radiating Level



ERL: Generally around 8-10 km

$E_{out} = \epsilon \sigma T_E^4$

Effective Radiating Level

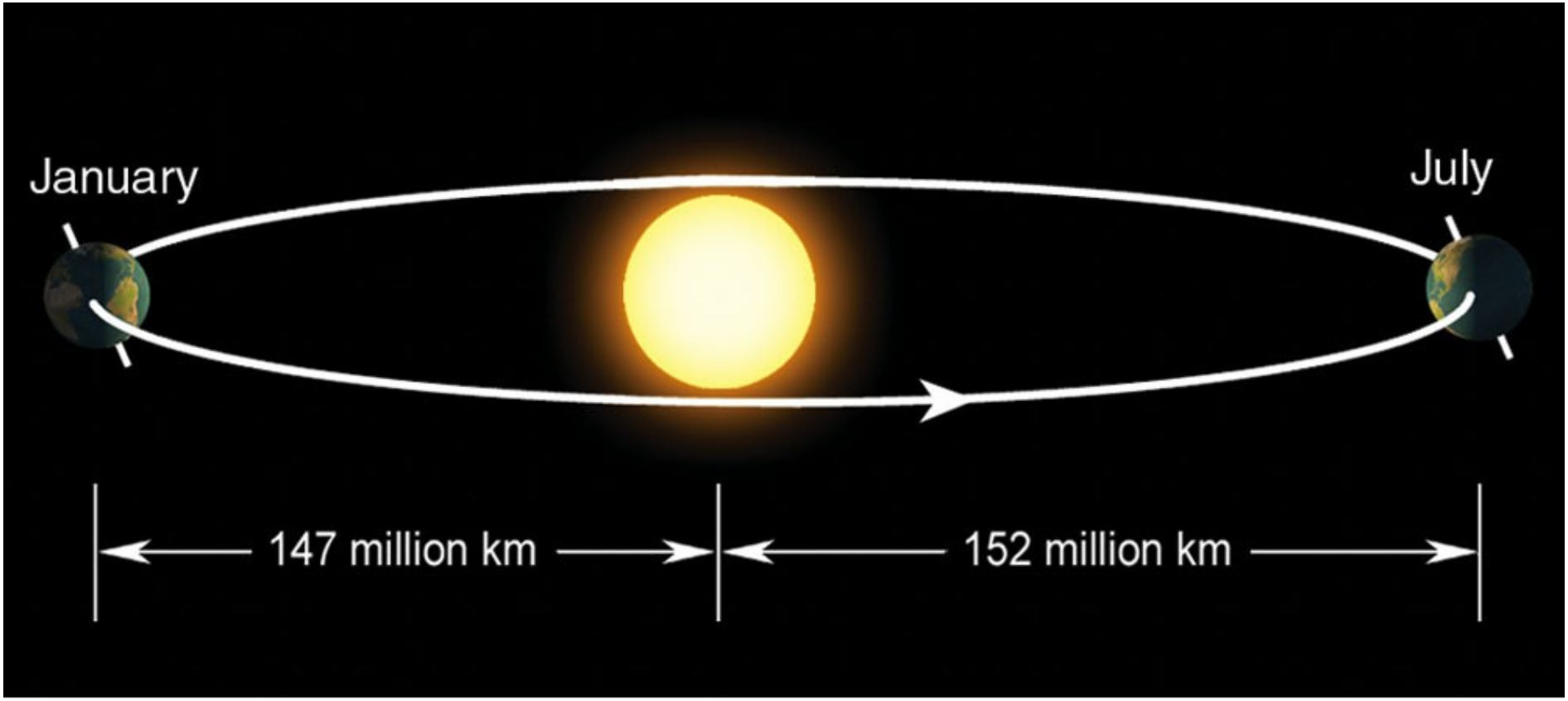


ERL: Generally around 8-10 km

The Seasons

Why isn't it always Fall?

Distance from the Sun



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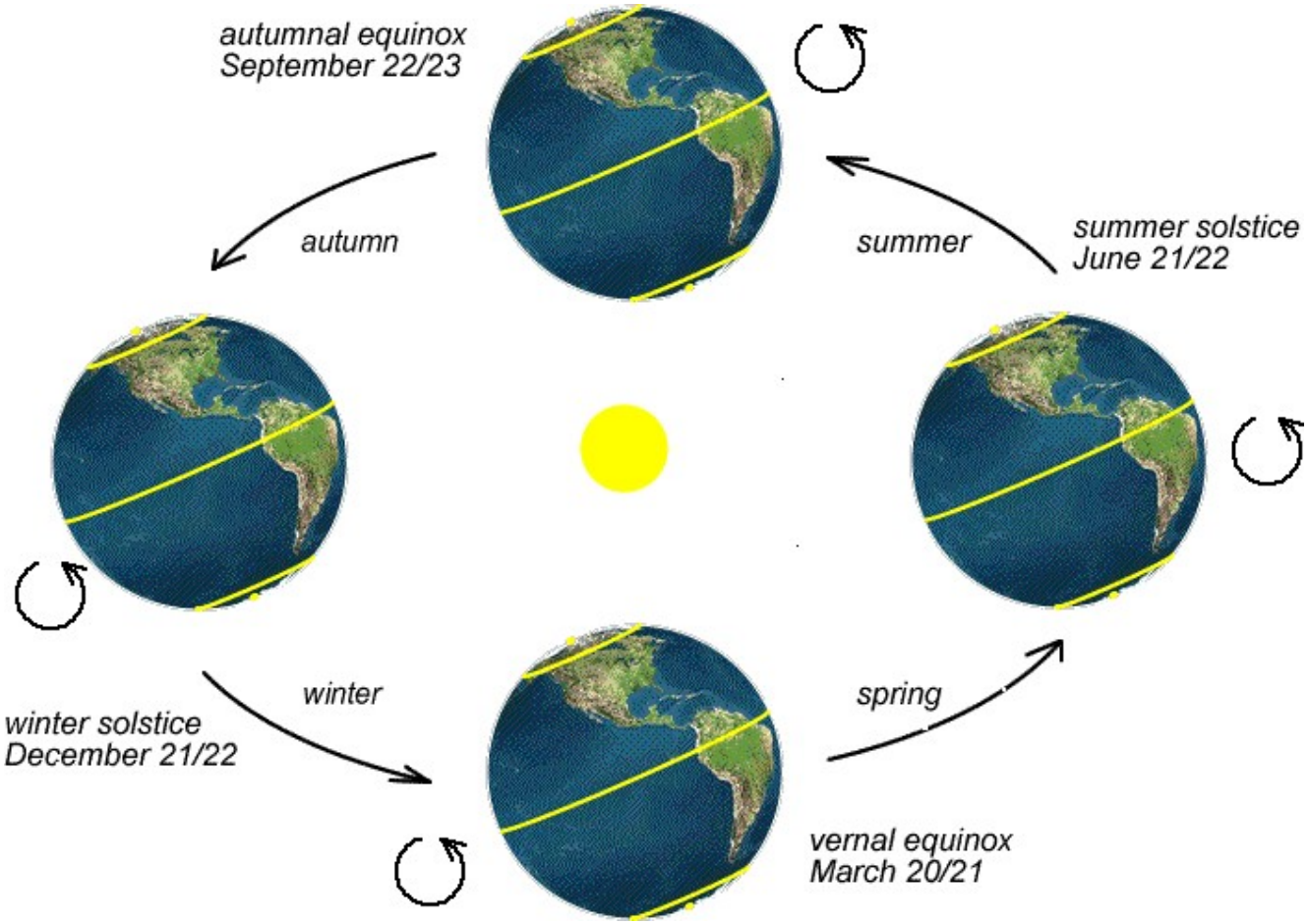
Northern hemisphere is closest in the winter!

Seasons



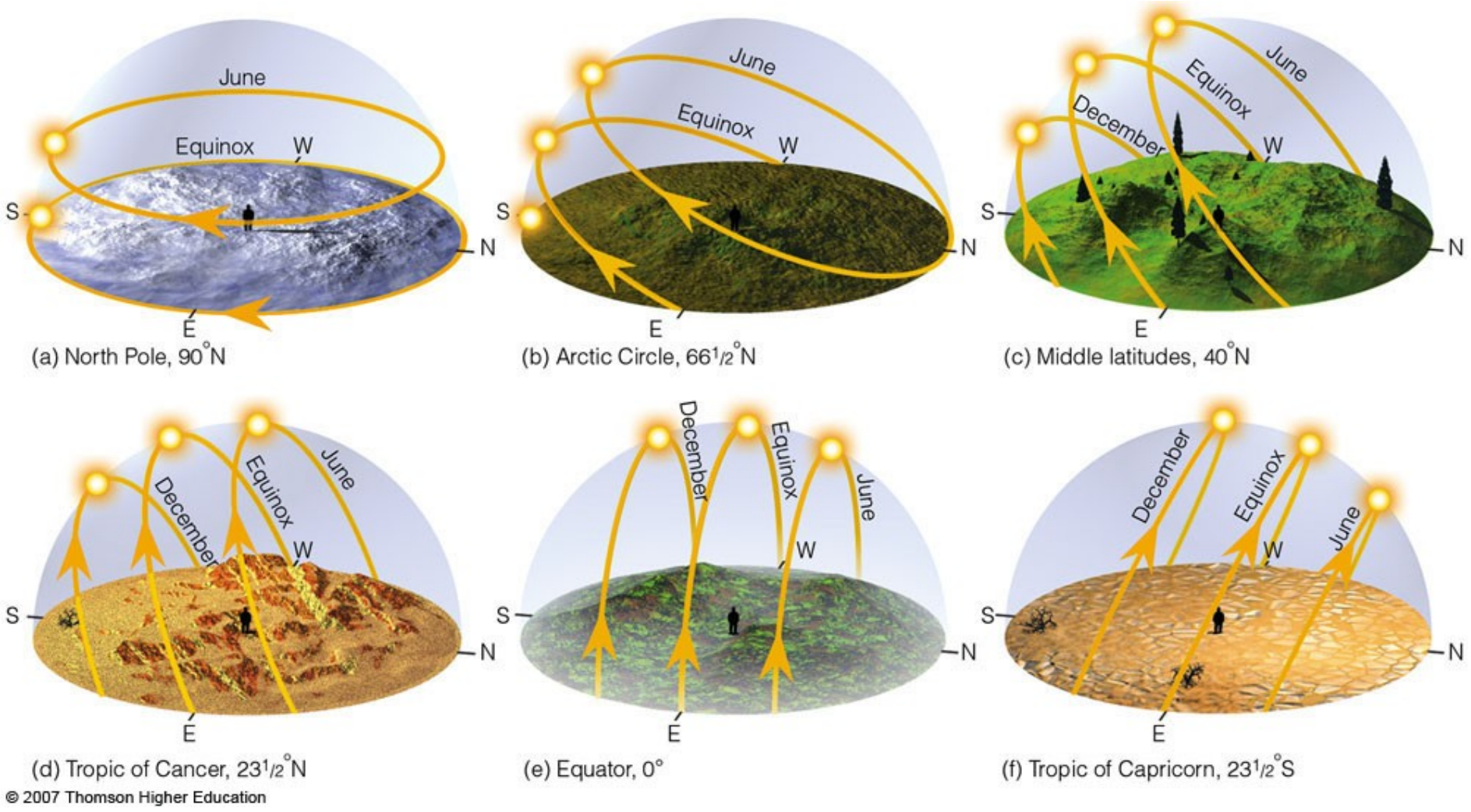
The Earth's spin axis maintains its orientation

22 Dec 2022

NOAA GOES-East

Data: NOAA, Processing: @simon_sat / NCEO

Heating from the Sun



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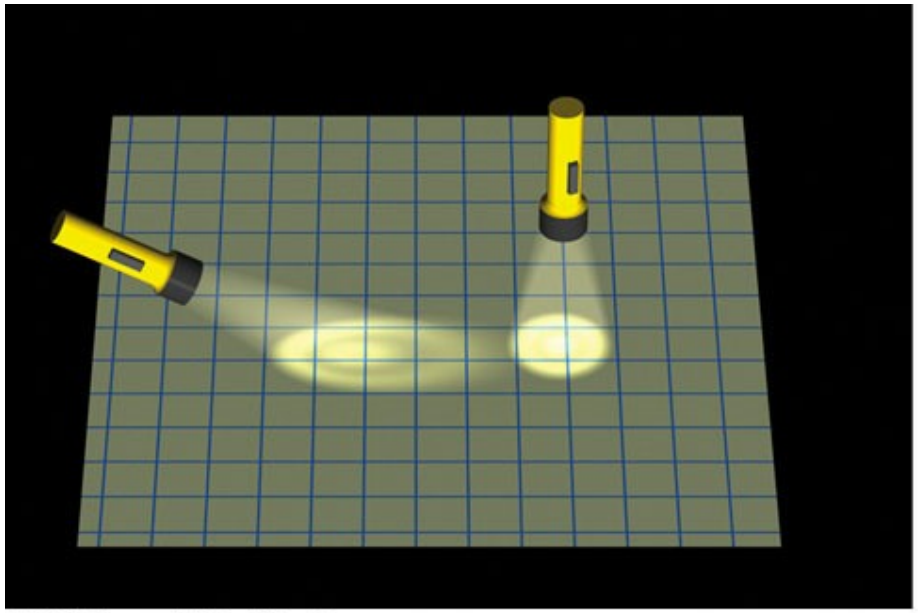
The sun's angle changes with time of day, with latitude, and with season.

Heating from the Sun



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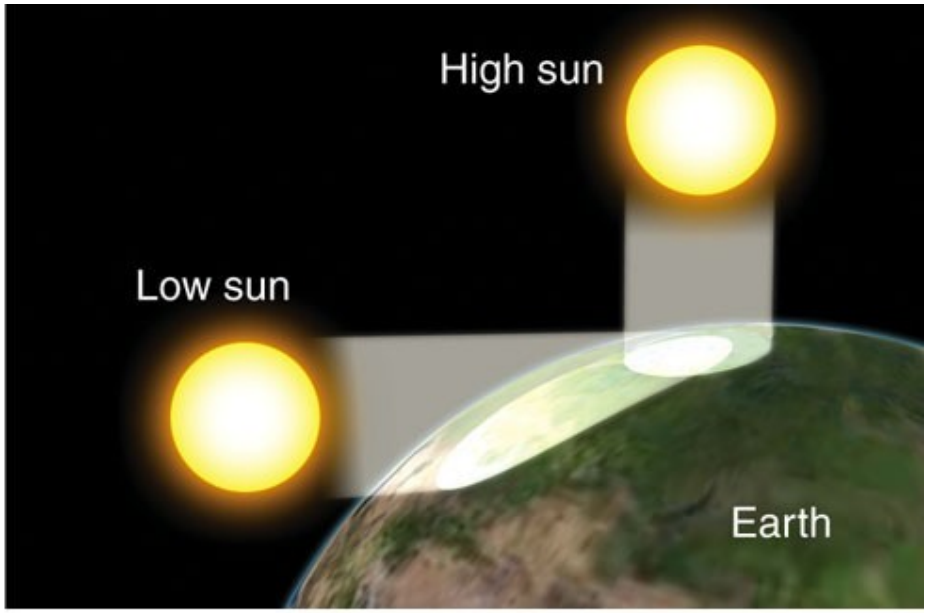
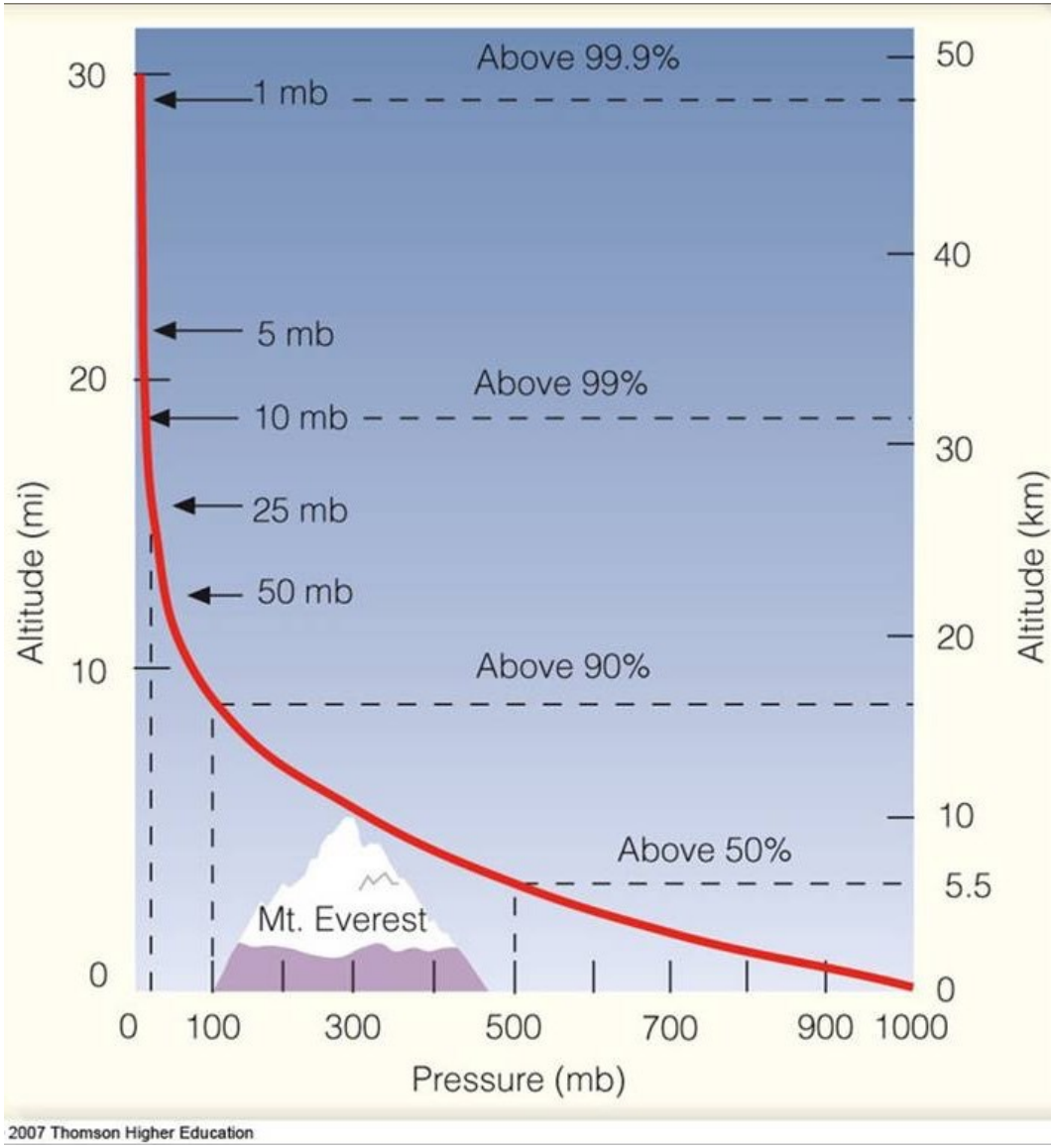


The sun's angle changes with time of day, with latitude, and with season.

Vertical Structure of the Atmosphere

Most of the air in the atmosphere lies within a few tens of kilometres of the surface.
Half of it lies within 5.5 km, 99.9% within 30 km.
Contrast with:
Earth radius: 6371 km
McGill to the
Olympic Stadium: 7 km

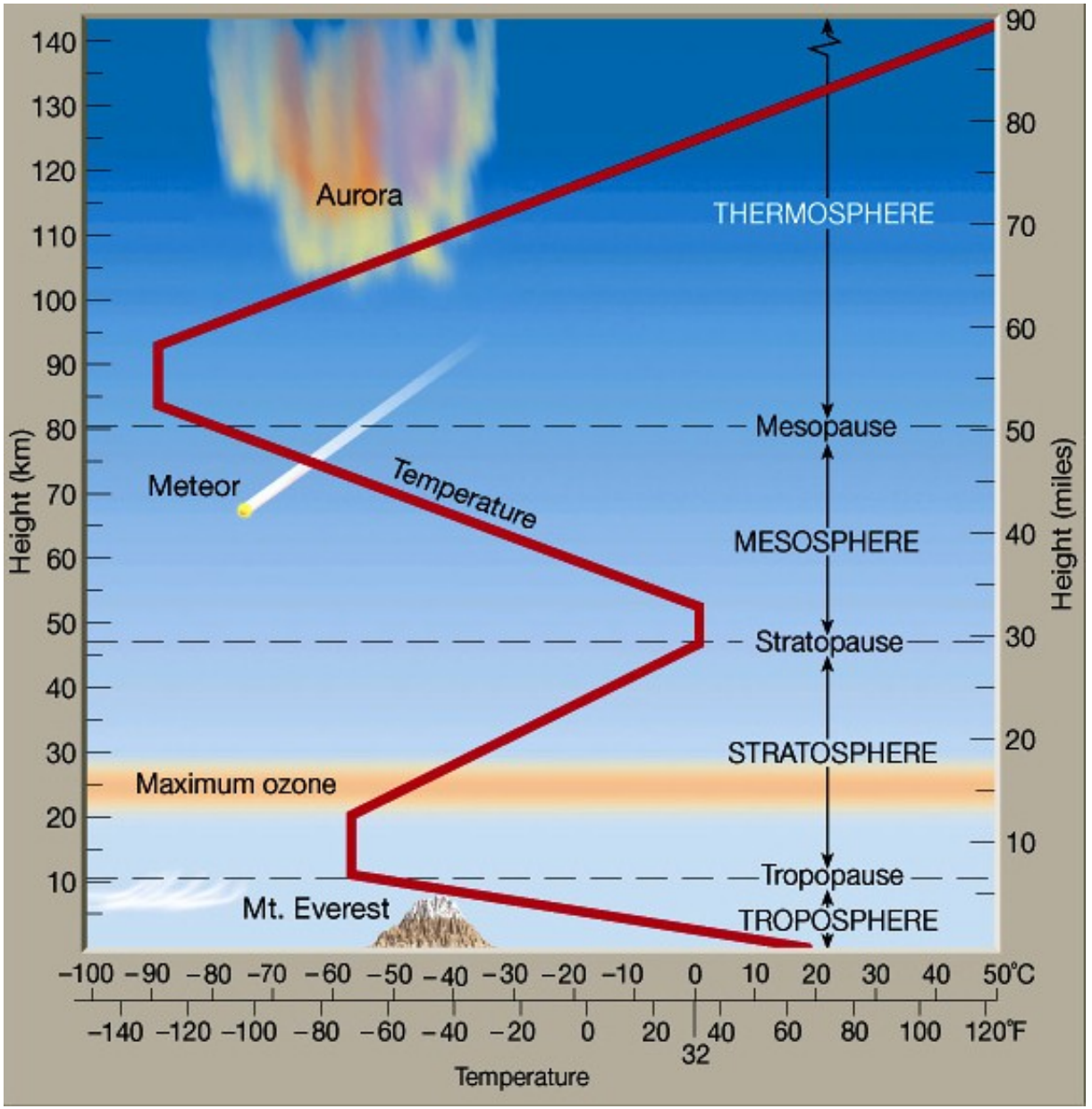


Layered Structure of the Atmosphere

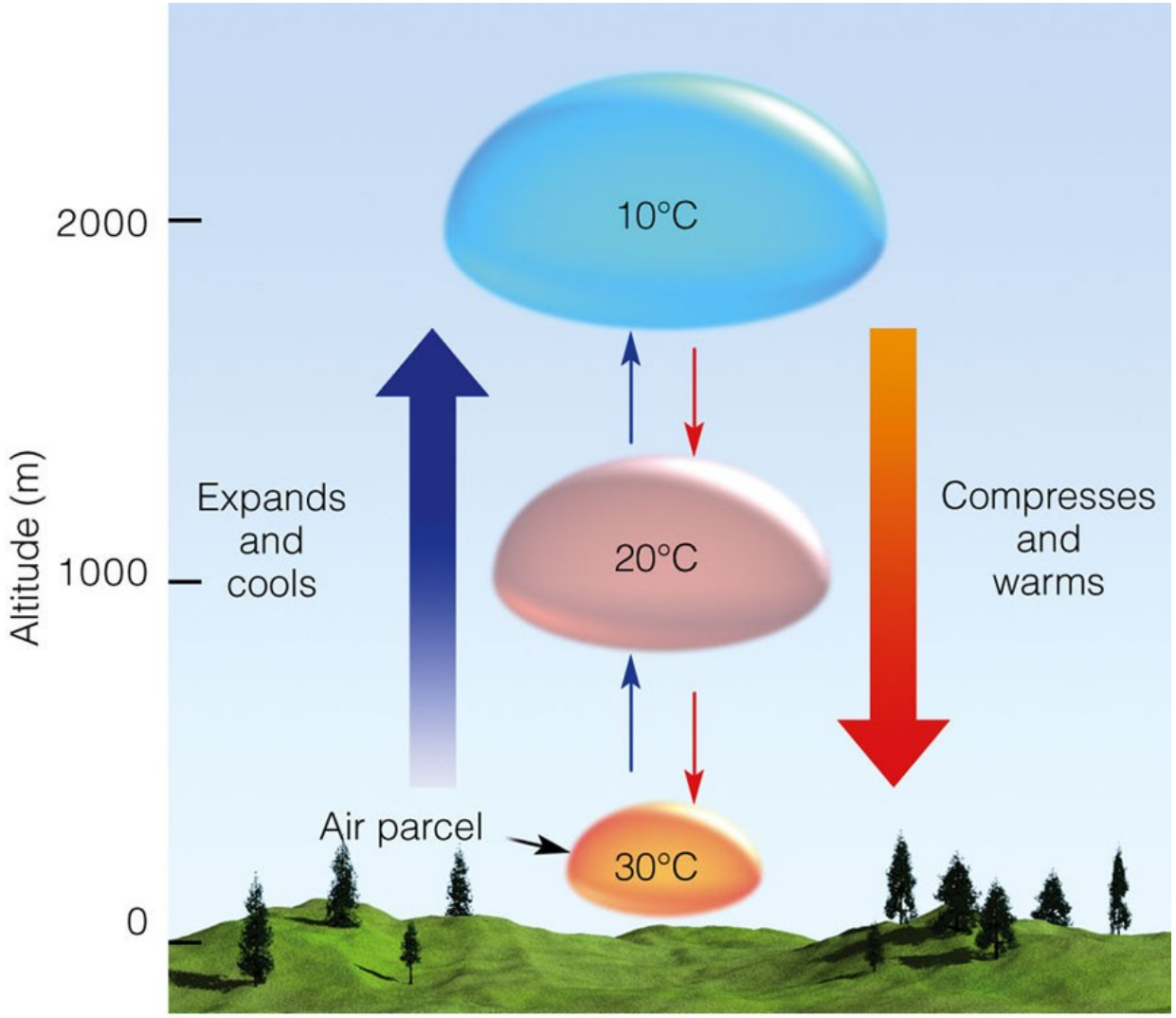
Layers are determined by how temperature changes with altitude.

The troposphere is where weather occurs (0 → 10-15 km);

Above lie the stratosphere, mesosphere and thermosphere.



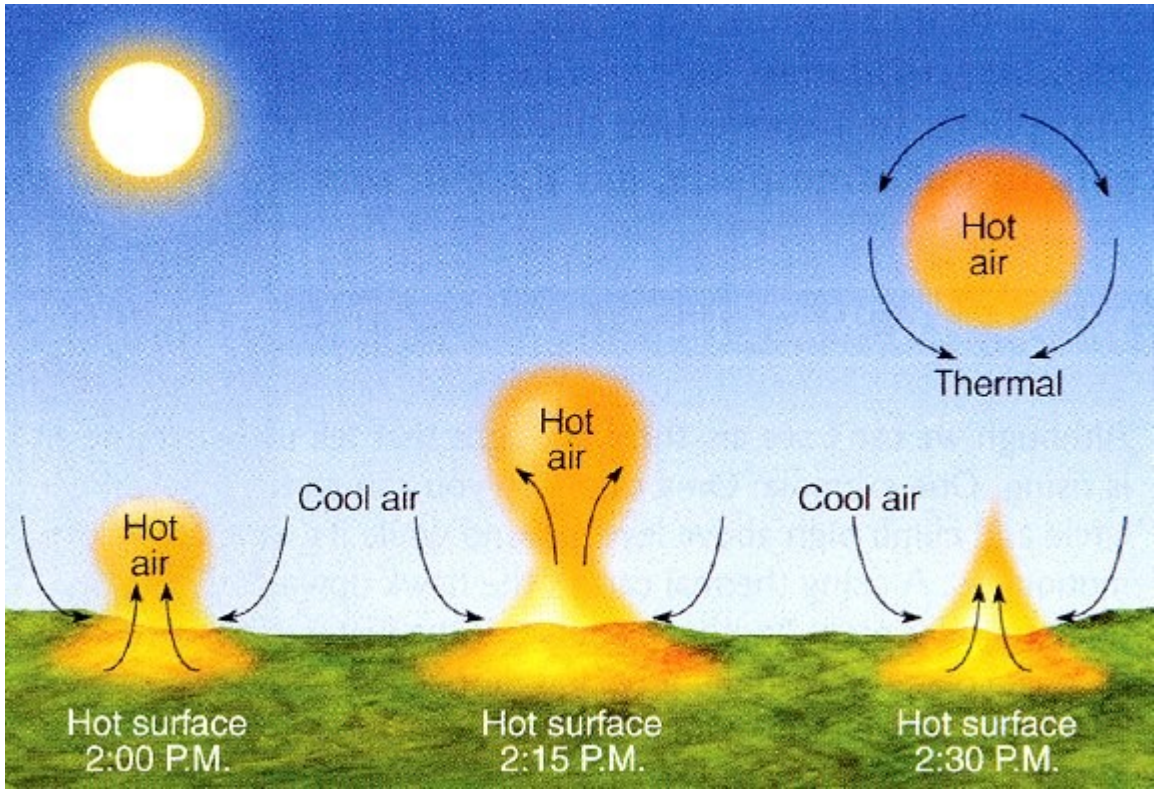
Air Density



Hot air is less dense than cold air: air expands and cools as it rises

Vertical Motion of Air

Yet, as long as it remains warmer than its immediate surroundings, warm air will keep going up.



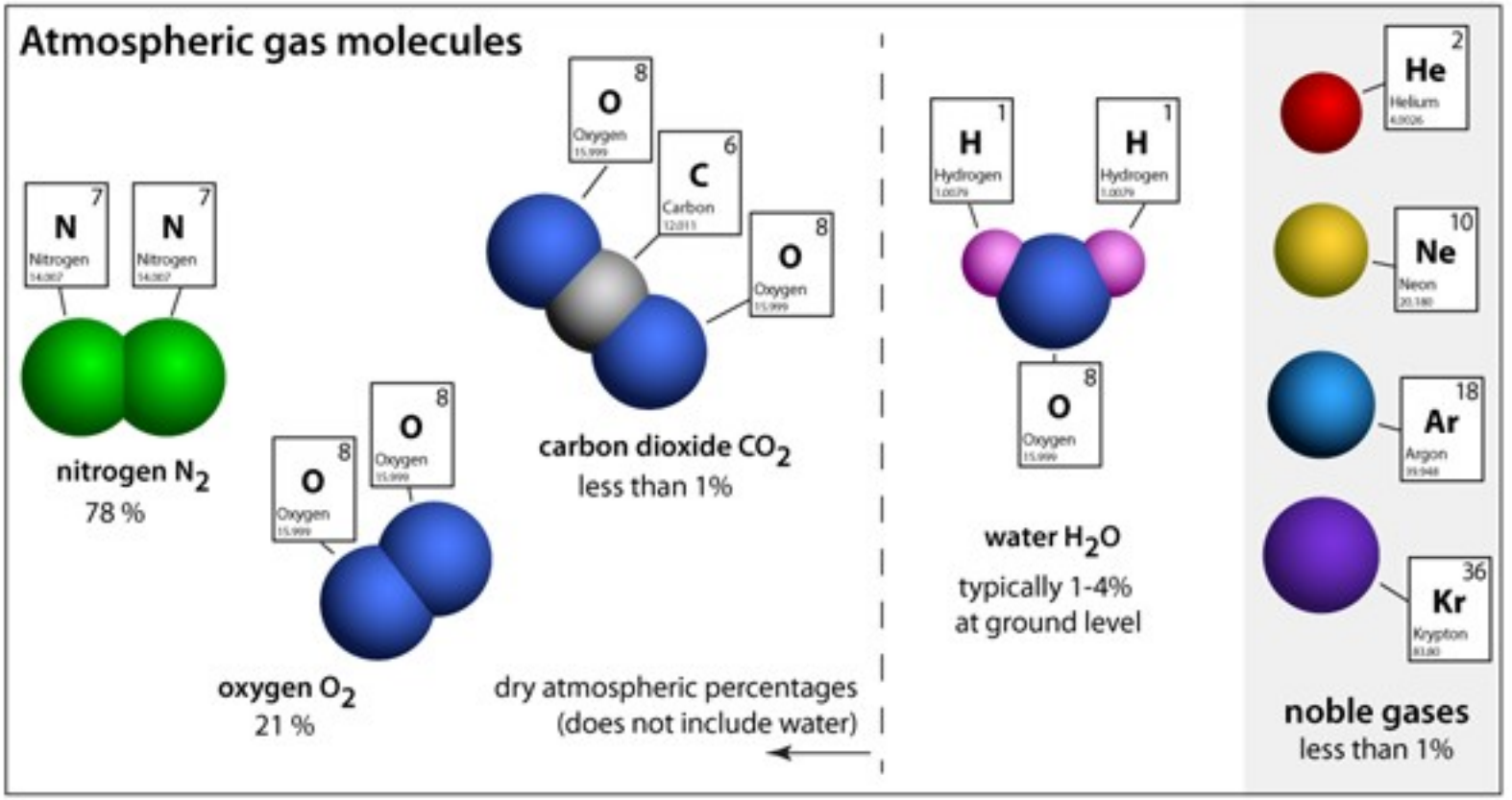
In parallel, as the amount of water that can stay in vapour form decreases as temperature decreases; when air becomes cold enough, condensation occurs.

Density of Air

Moist Air is less dense than dry air:

Molecular mass of N_2: 28

Molecular mass of H_2O: 18



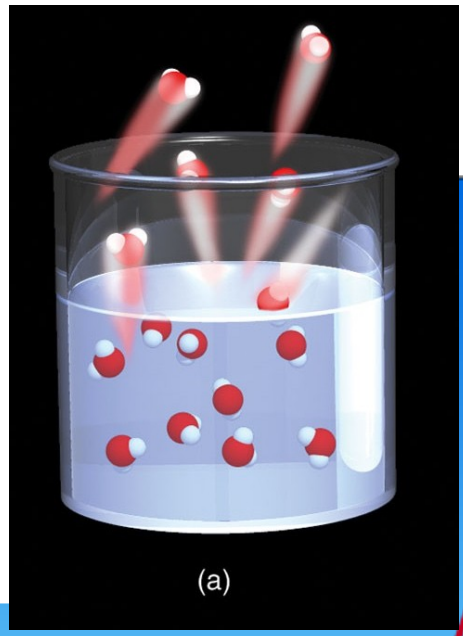
Saturation and Temperature

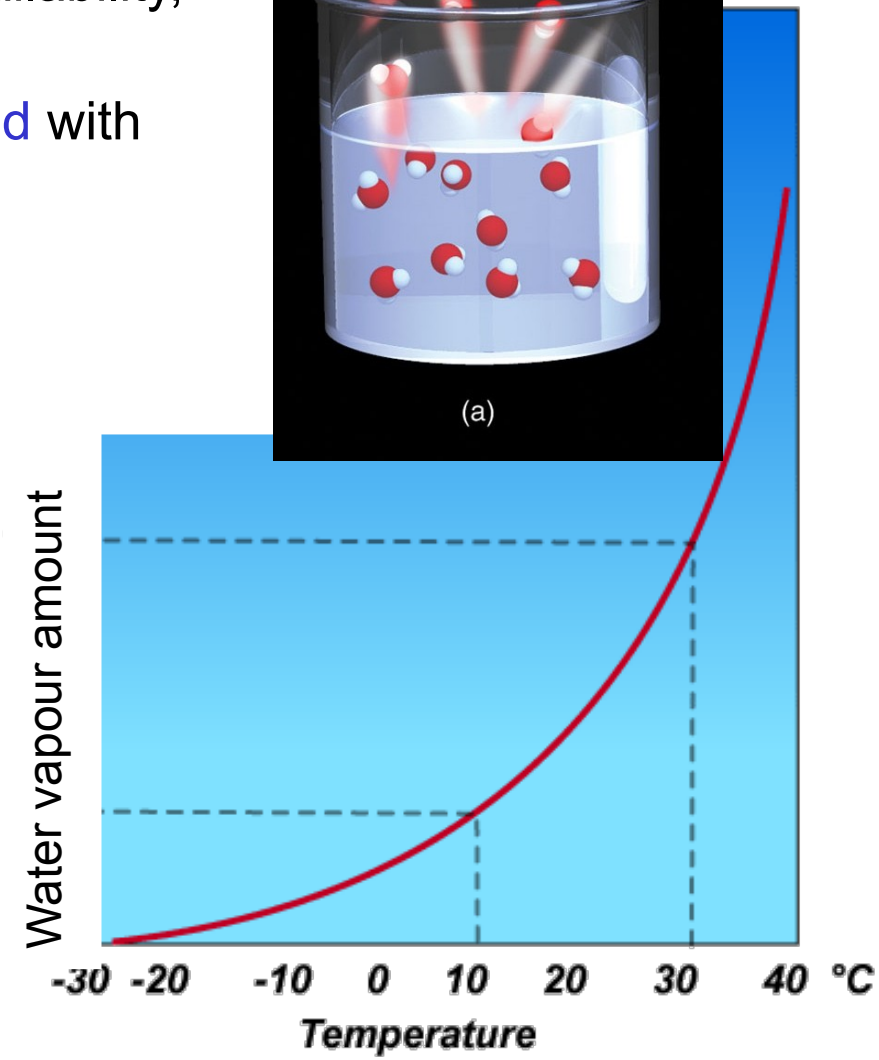
Rate of condensation is determined by the amount of water vapour;

Rate of evaporation is linked to energy availability, or temperature.

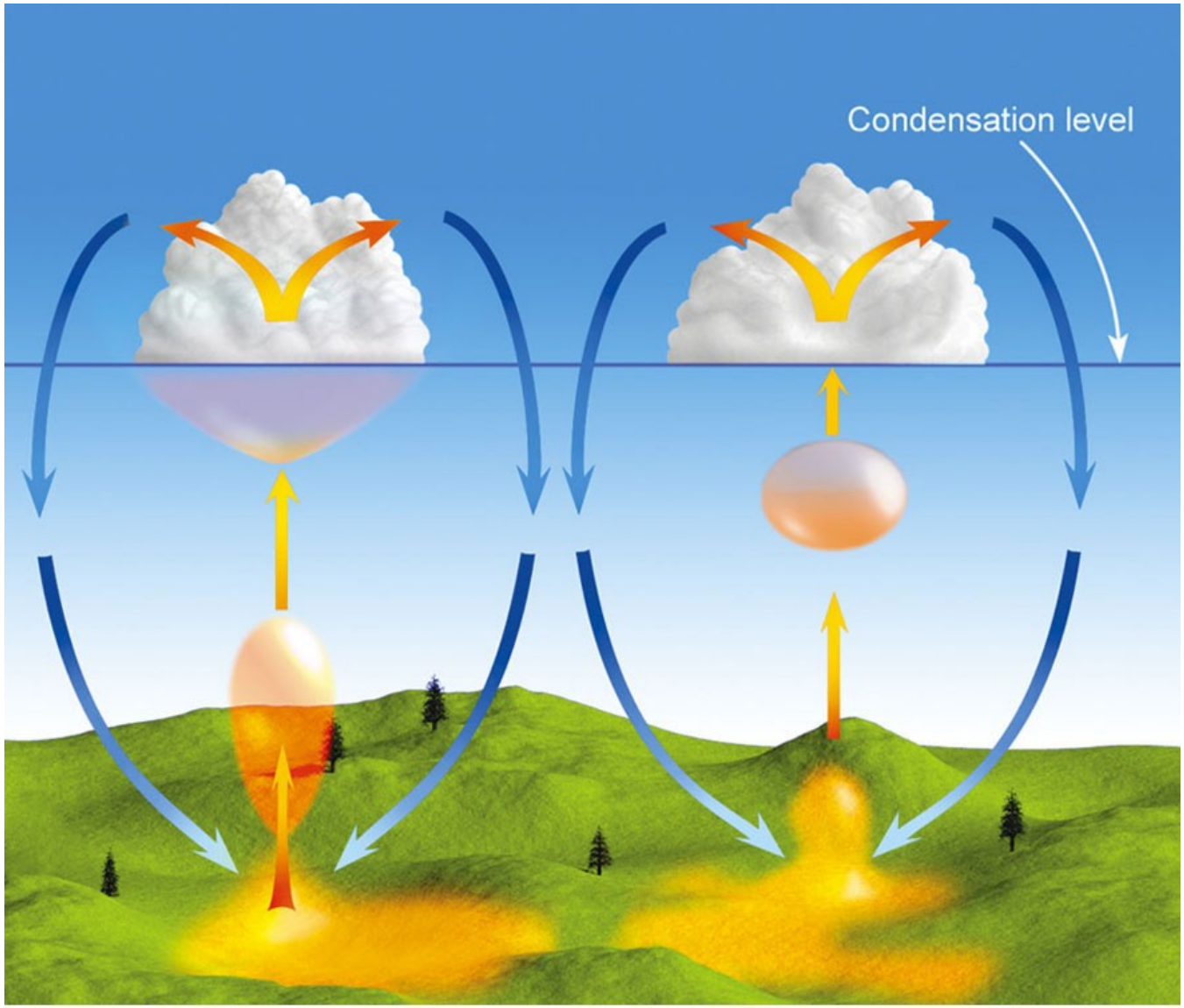
When the two are balanced, air is saturated with vapour (100% humidity).

When saturated air cools, net condensation occurs on (almost) any solid or liquid surface, including dust particles → clouds form.





Surface Heating; Cooling Aloft



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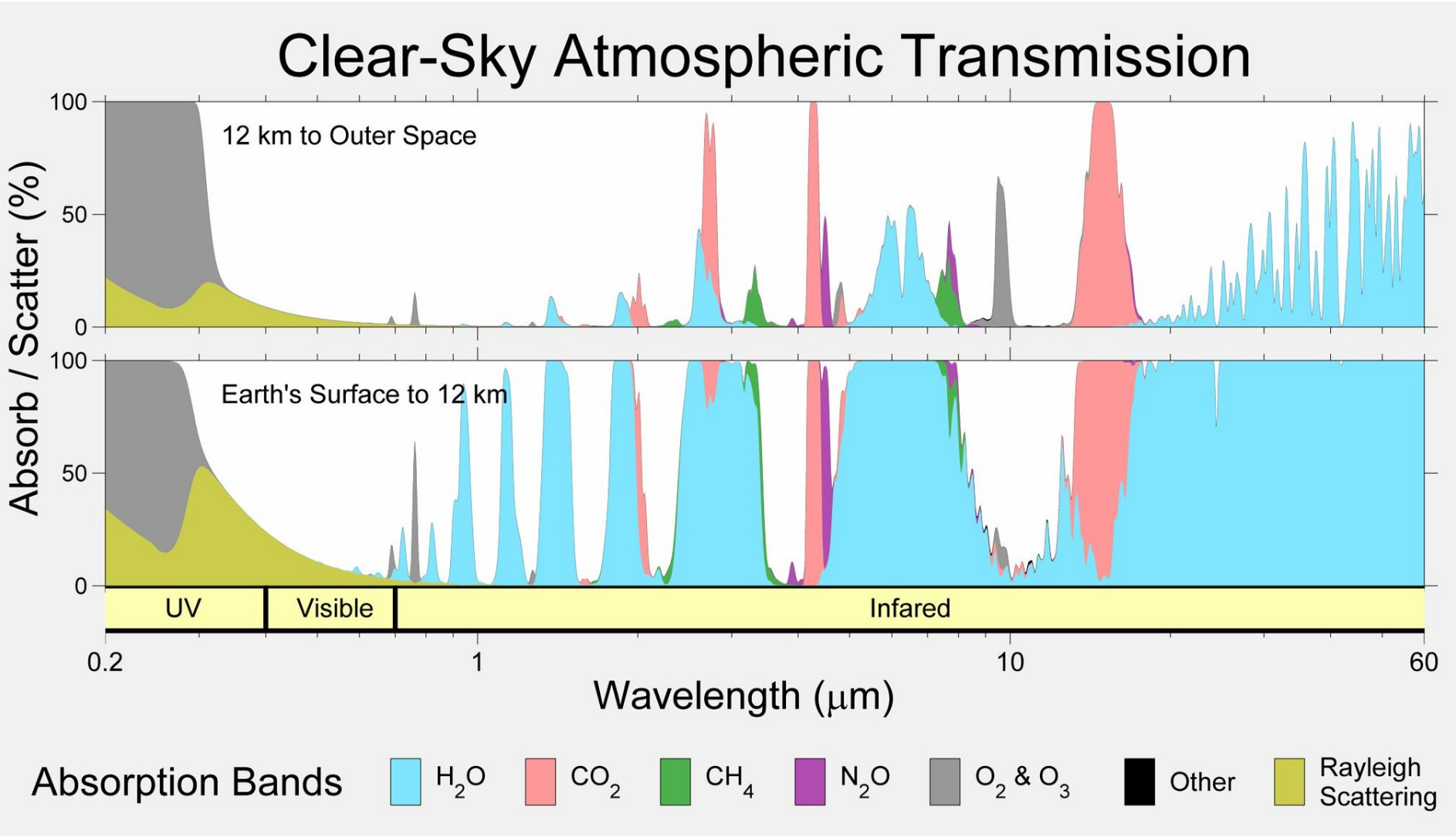
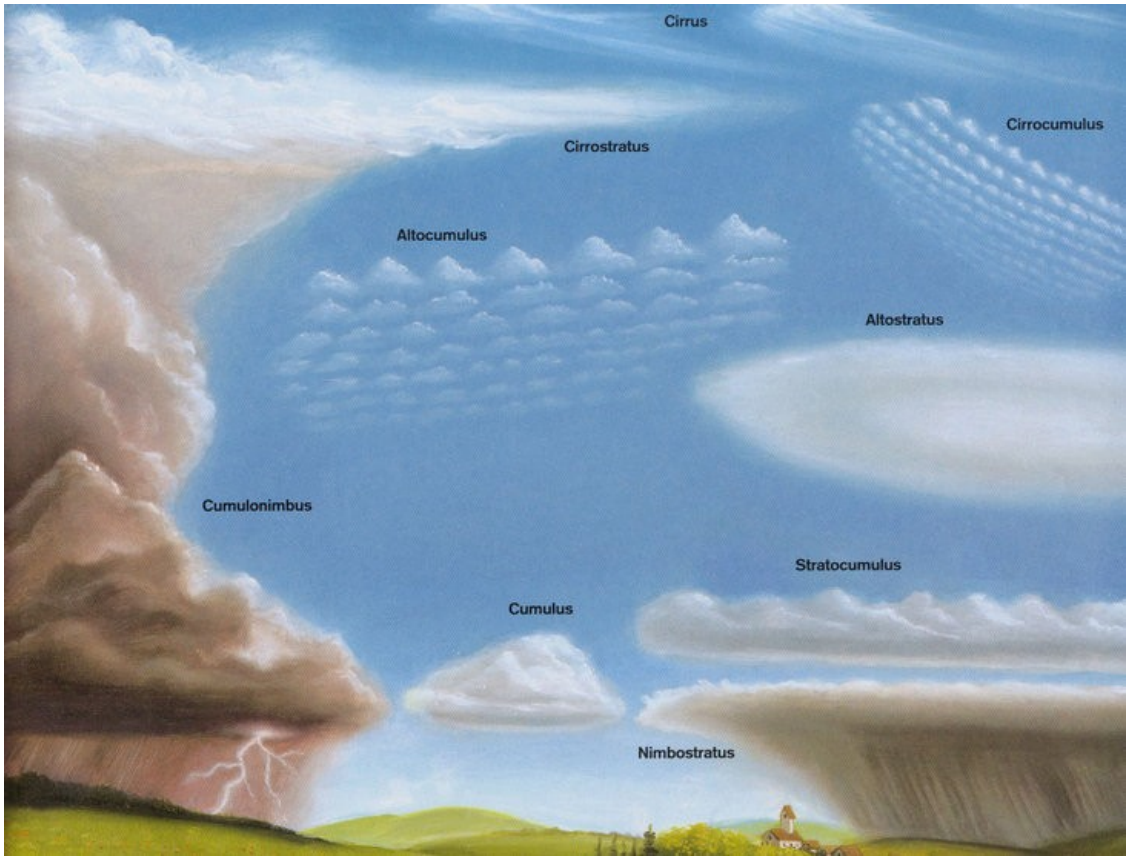
Warm land heats the air above it


Clear-Sky Atmospheric Transmission
12 km to Outer Space
Earth's Surface to 12 km
Absorb / Scatter (%)
100
50
0
UV
Visible
Infared
0.2
1
10
60
Wavelength (μm)
Absorption Bands
H2O
CO2
CH4
N2O
O2 & O3
Other
Rayleigh Scattering

R A Rohdes

Clouds and Vertical Motion



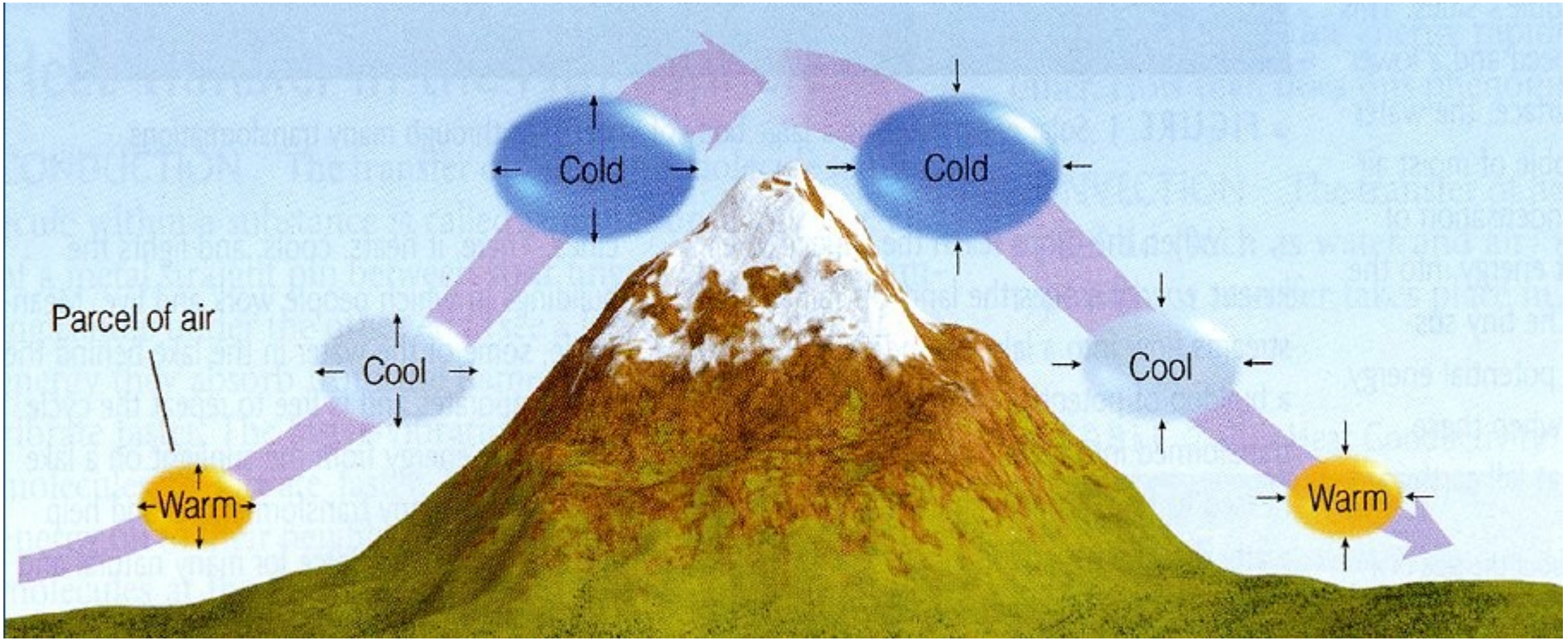
- Almost all clouds are formed by cooling due to rising of moist air, and subsequent condensation;
- Descending motion brings drier air from aloft and tends to suppress cloud formation.

Vertical Motion of Air: Forced Convection

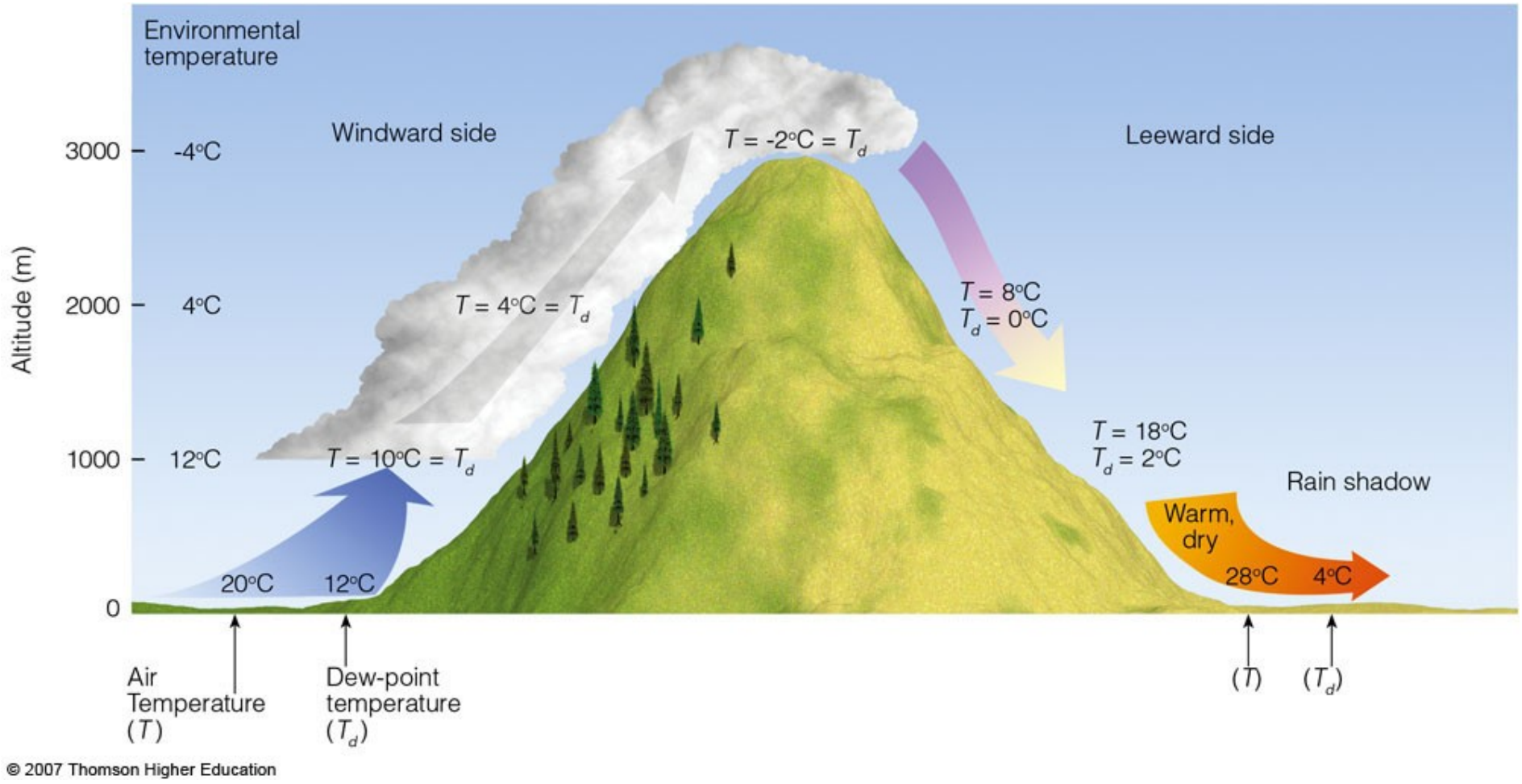
The real fun starts when air moves up:

Expanding air cools;

Air that expands cools + air that goes up expands = air that goes up cools.



Orographic Condensation



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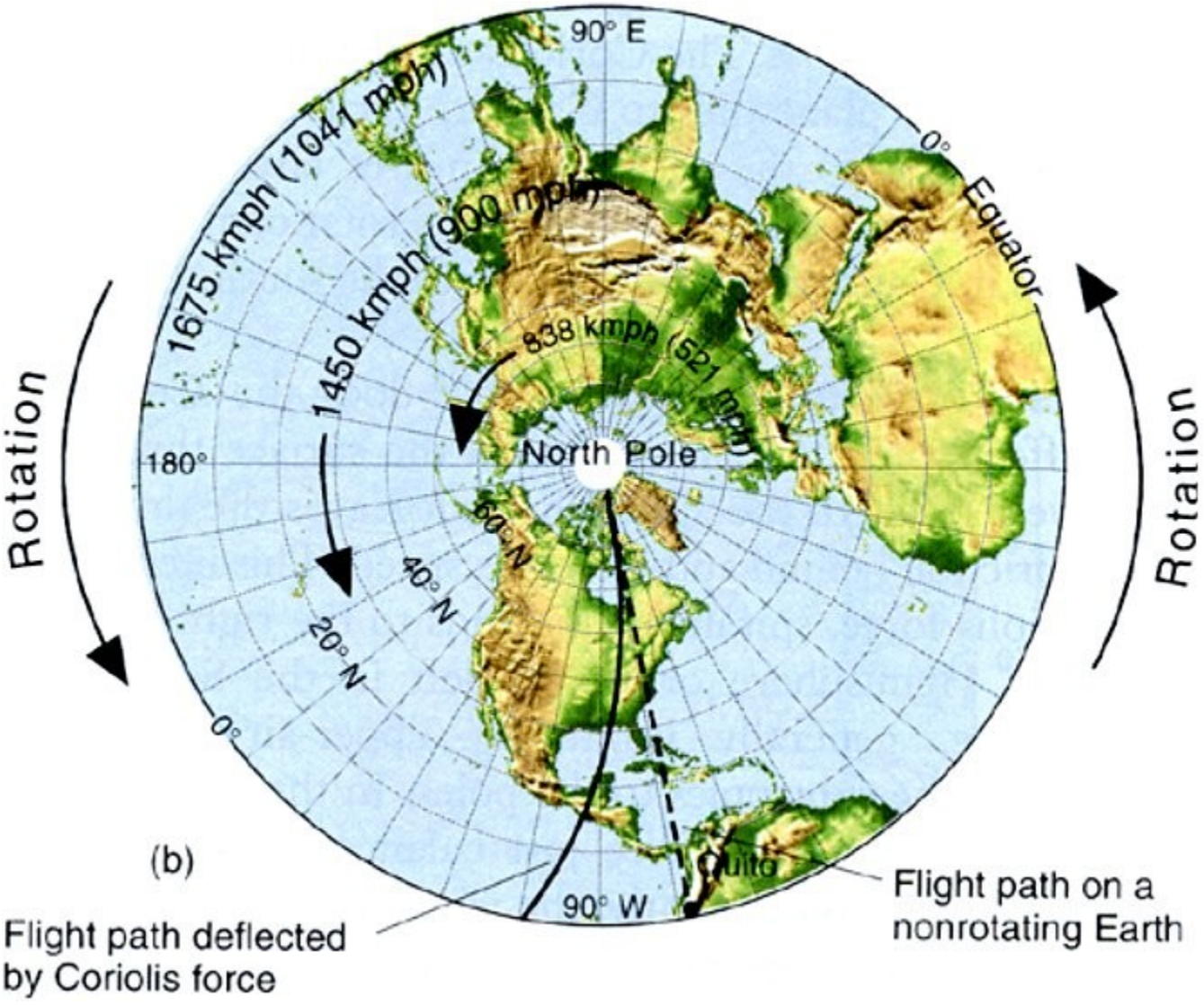
Dew point: the temperature of water vapor saturation

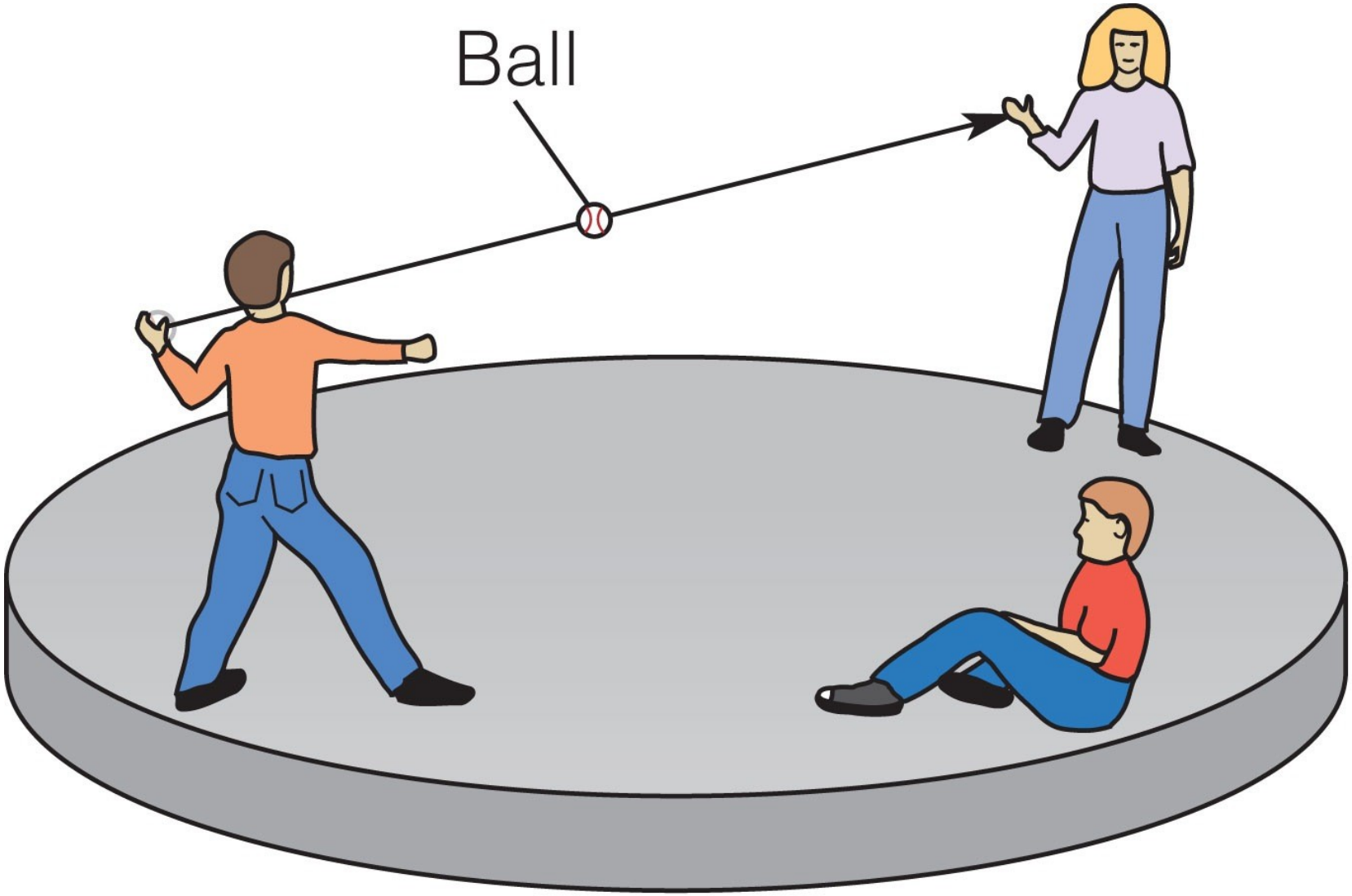
Cool, lower pressure air holds less moisture

Further Twist: The Coriolis Effect



Because we are on a rotating Earth, straight paths appear curved to us, as if a “force” was pulling on moving objects. Trajectories curve to the right in the Northern Hemisphere, and to the left in the Southern.

Good thorough explanation of Coriolis effect: http://www.dvandom.com/coriolis/



(a)

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Figure 2-4a p27

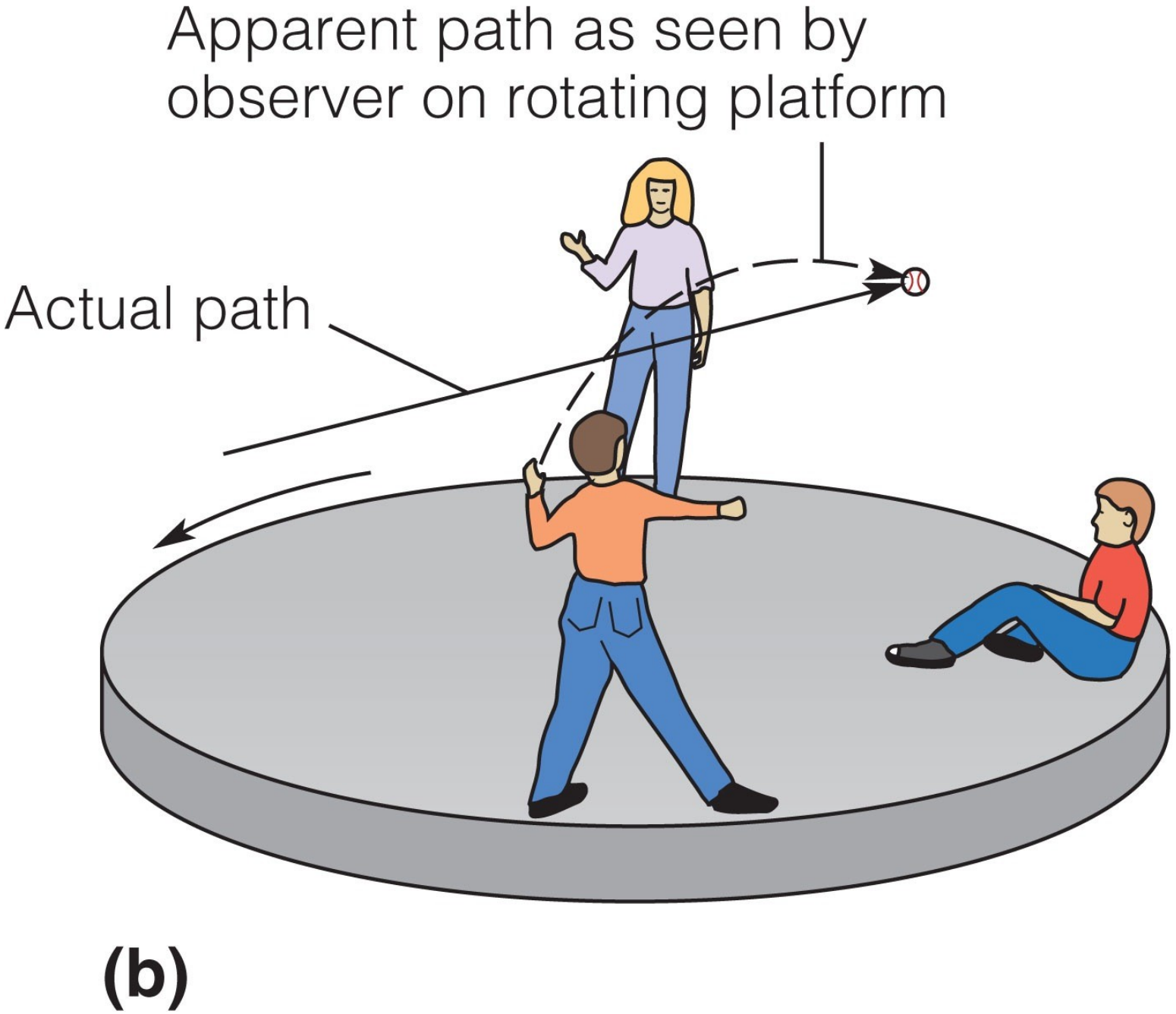


Figure 2-4b p27

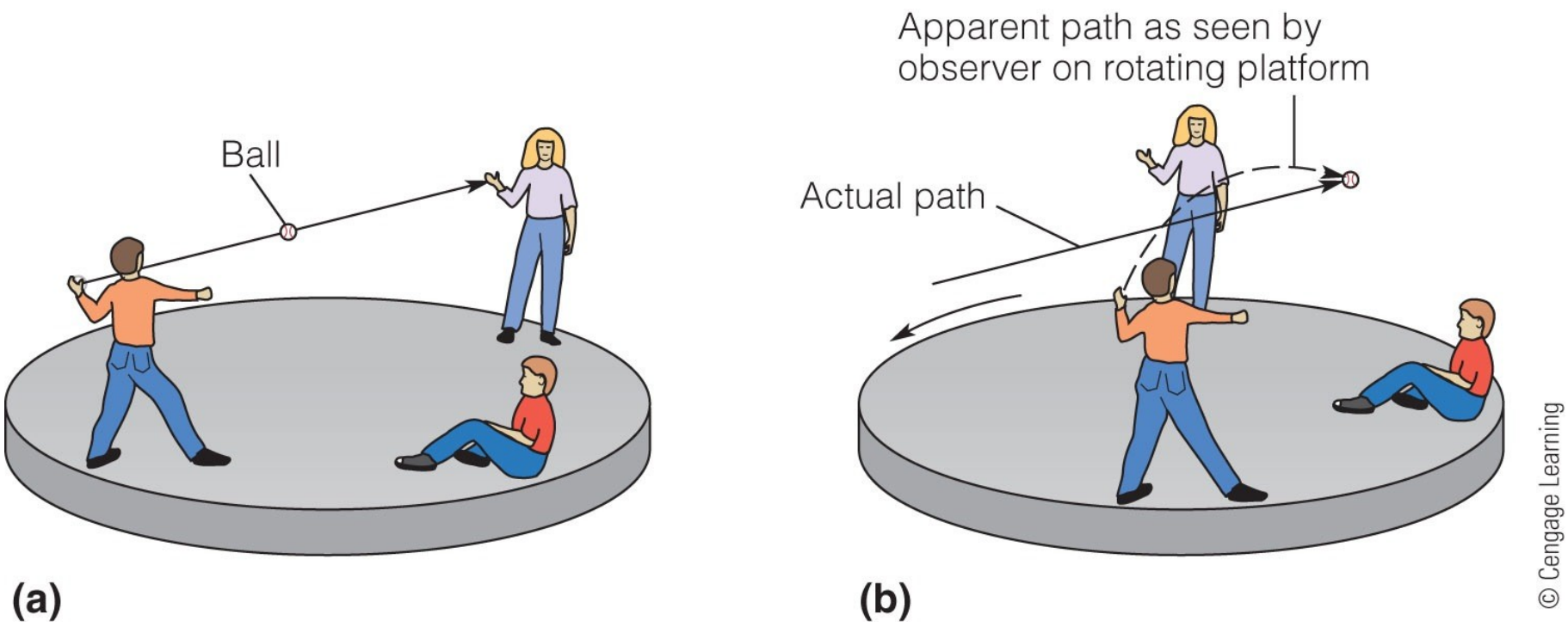
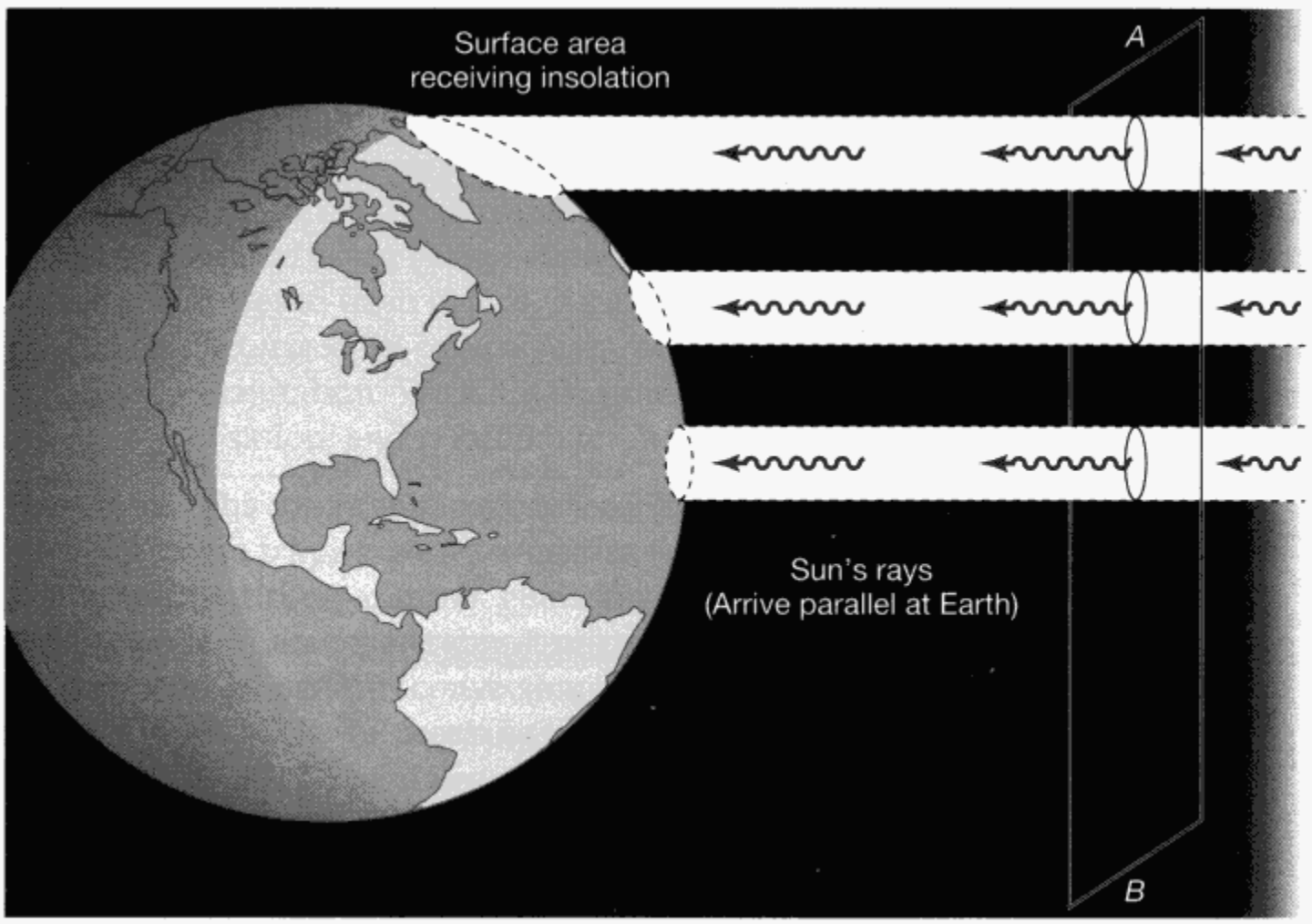


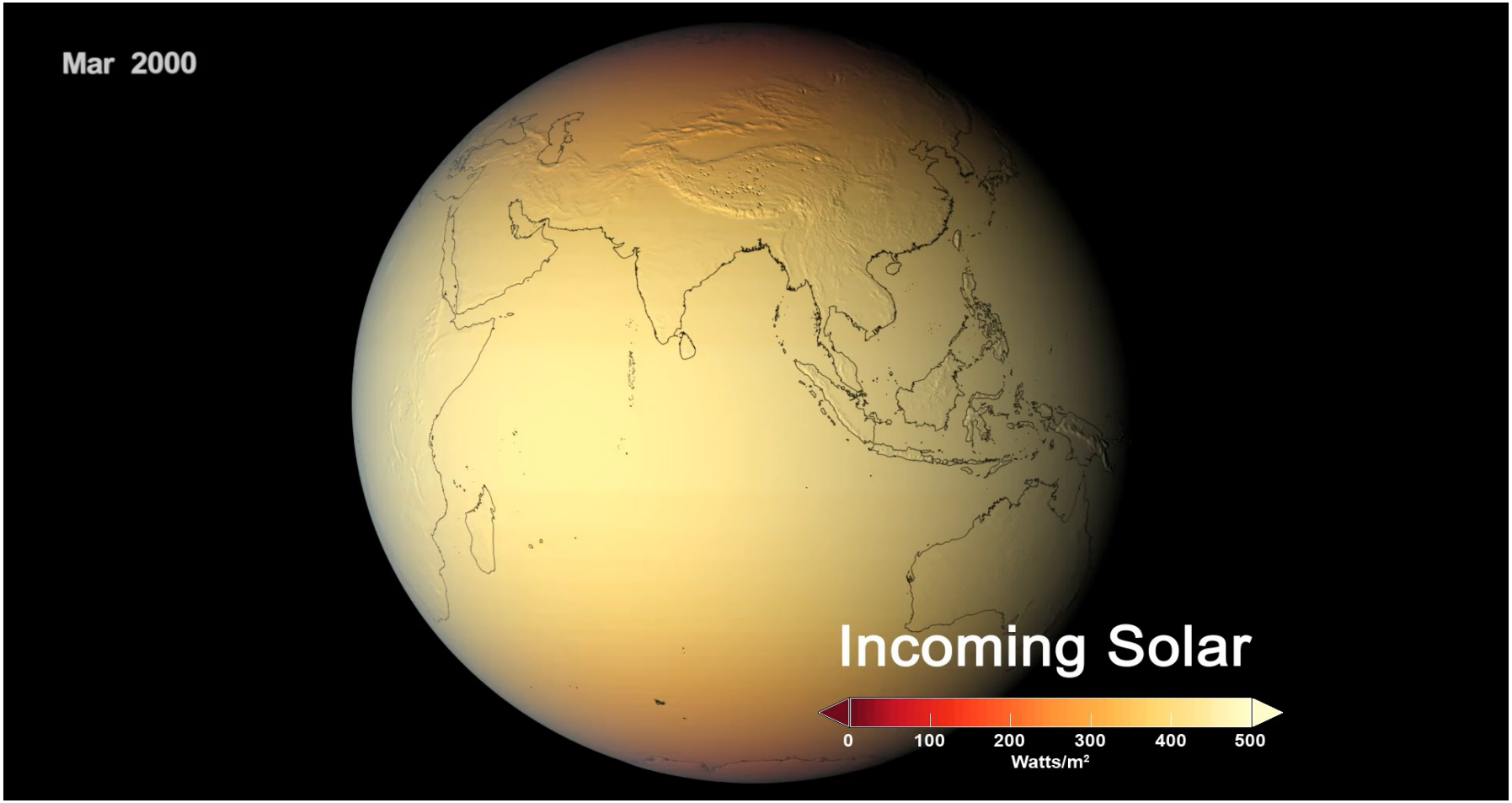
Figure 2-4 p27

Energy Distribution at the Surface



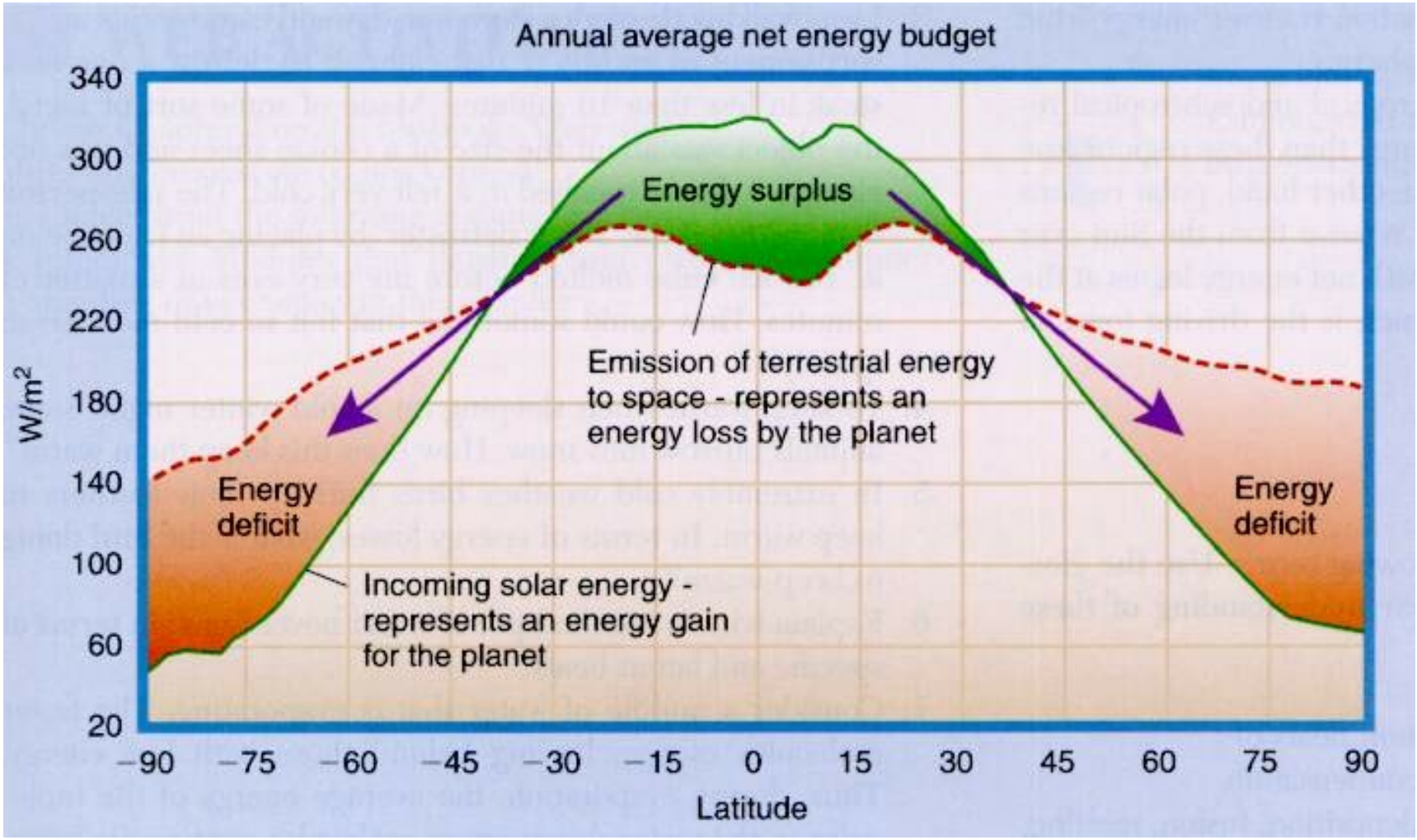
The equatorial latitudes receive more energy than the polar latitudes.

Uneven Heating from the Sun



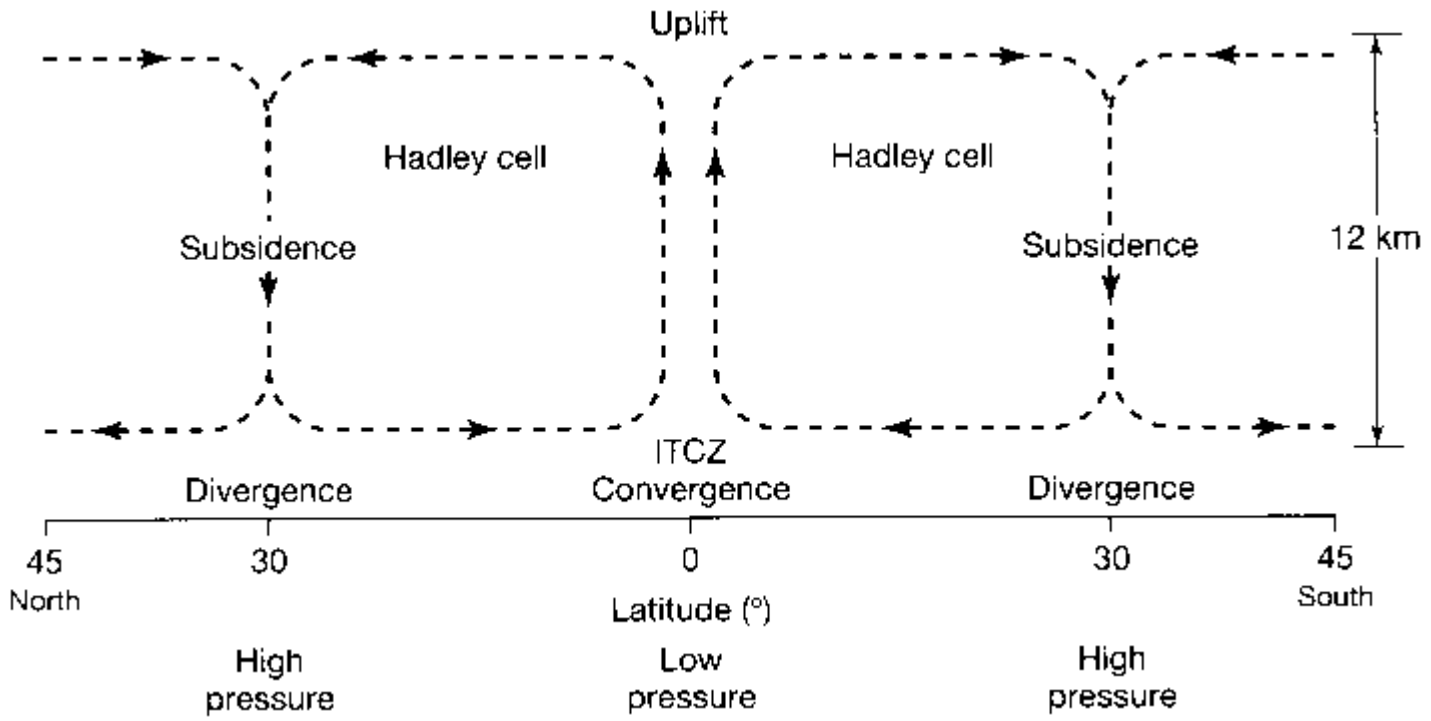
Weather is why we can’t use global averages!

Heating vs. Latitude



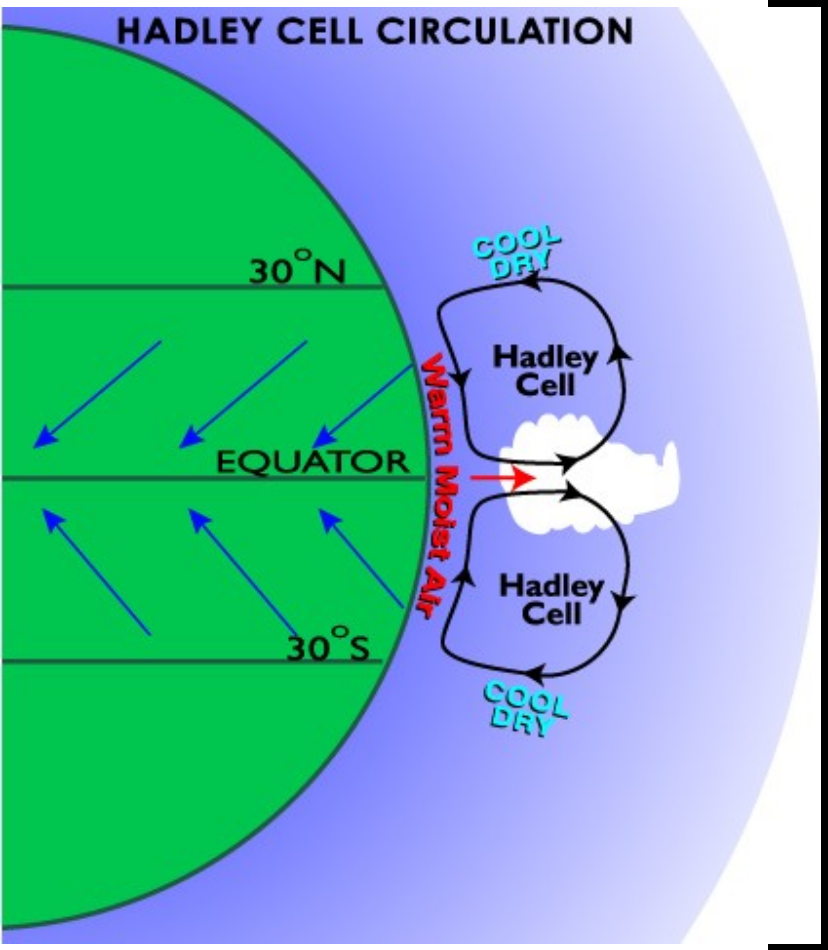
If this was the whole story, tropical regions should get gradually warmer and high latitudes colder. What's missing?
The transport of heat by the atmosphere and oceans.

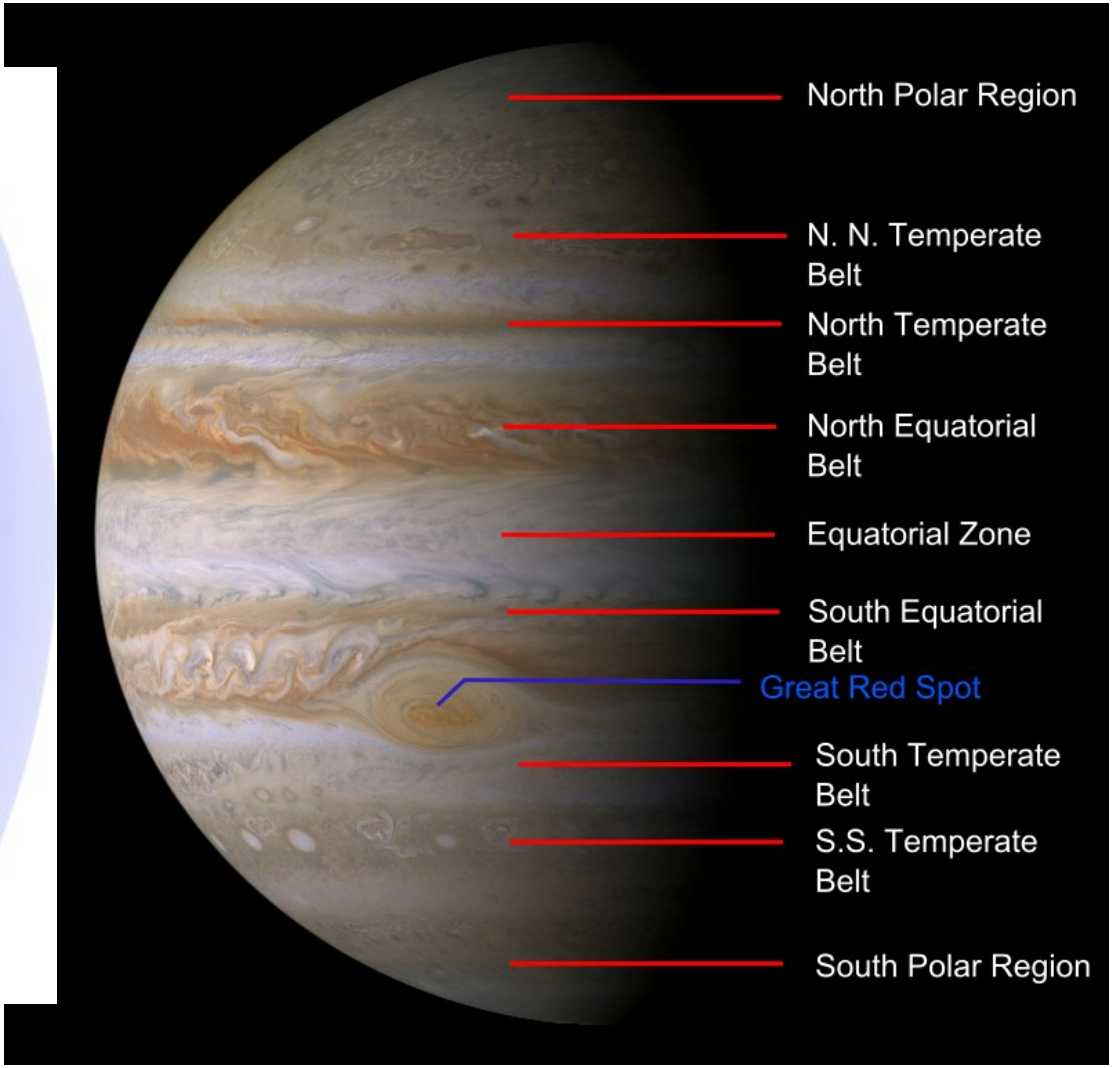
General Circulation – Tropics



- Warm moist equatorial air rises and moves towards the poles (Hadley cell), leading to cloud formation;
- There is sinking motion at about 30° latitude in both hemispheres → Sunny and dry areas

Hadley Cells





Jupiter rotates much faster: more cells than Earth

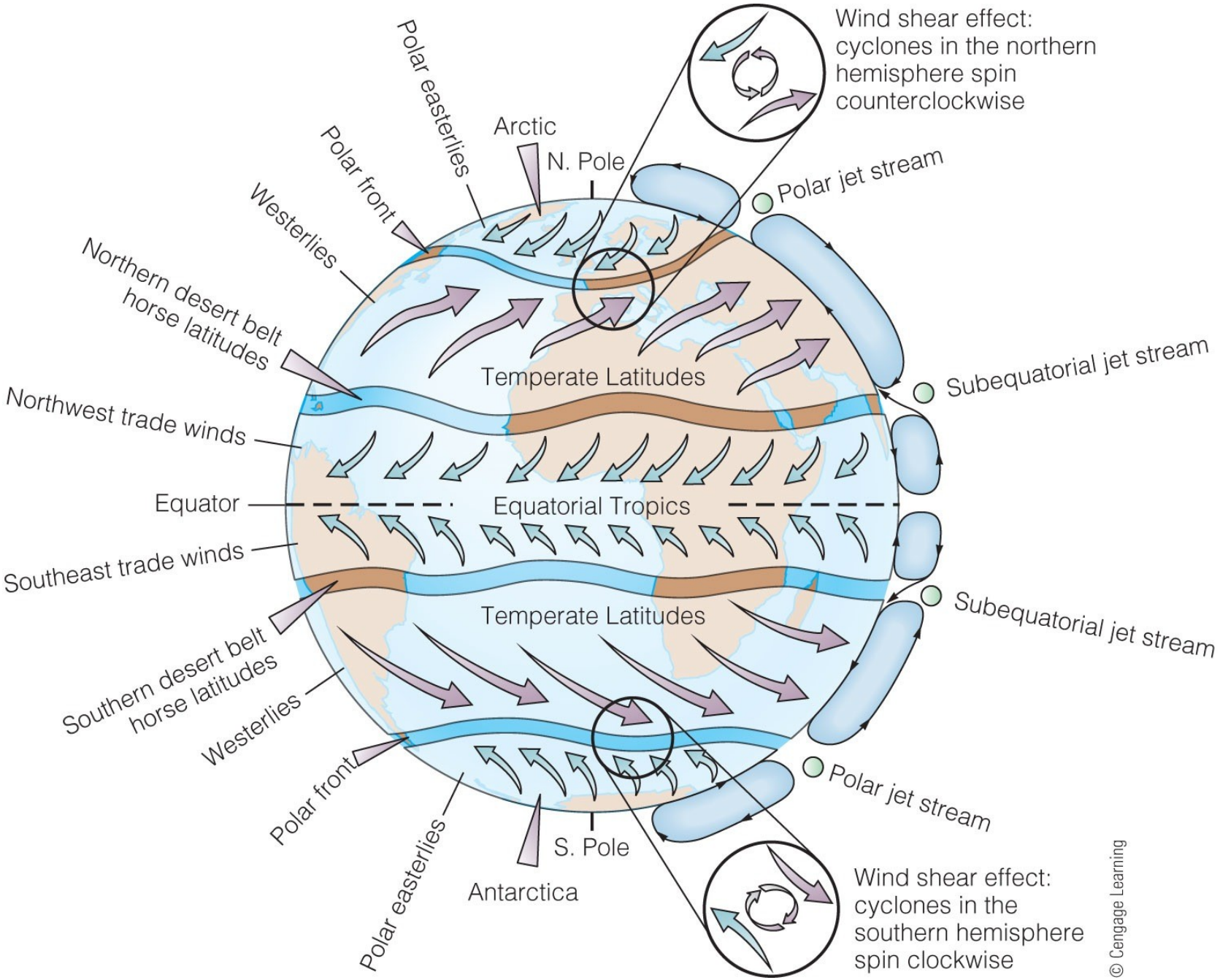
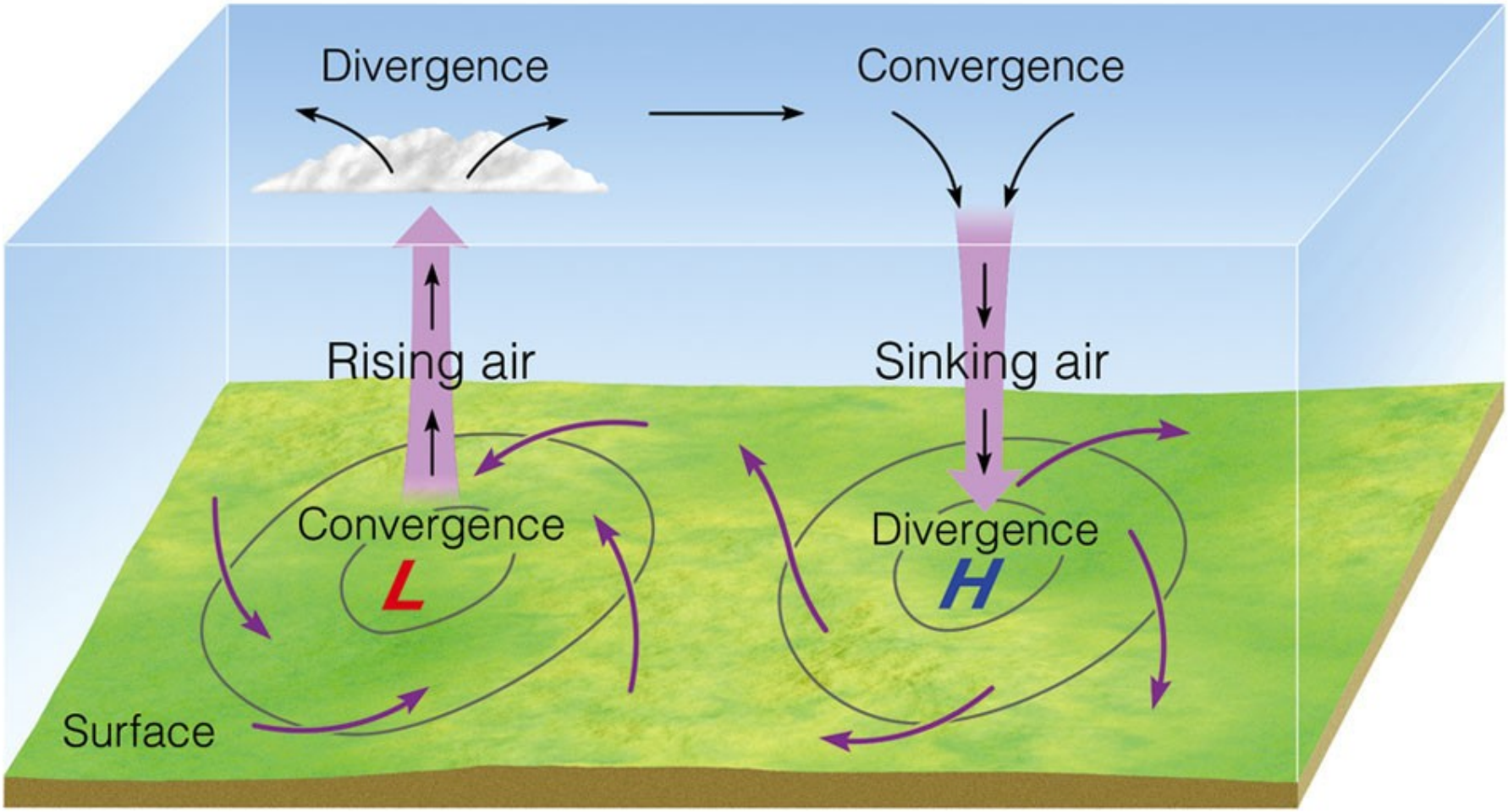


Figure 2-5 p27

Convection and circulation



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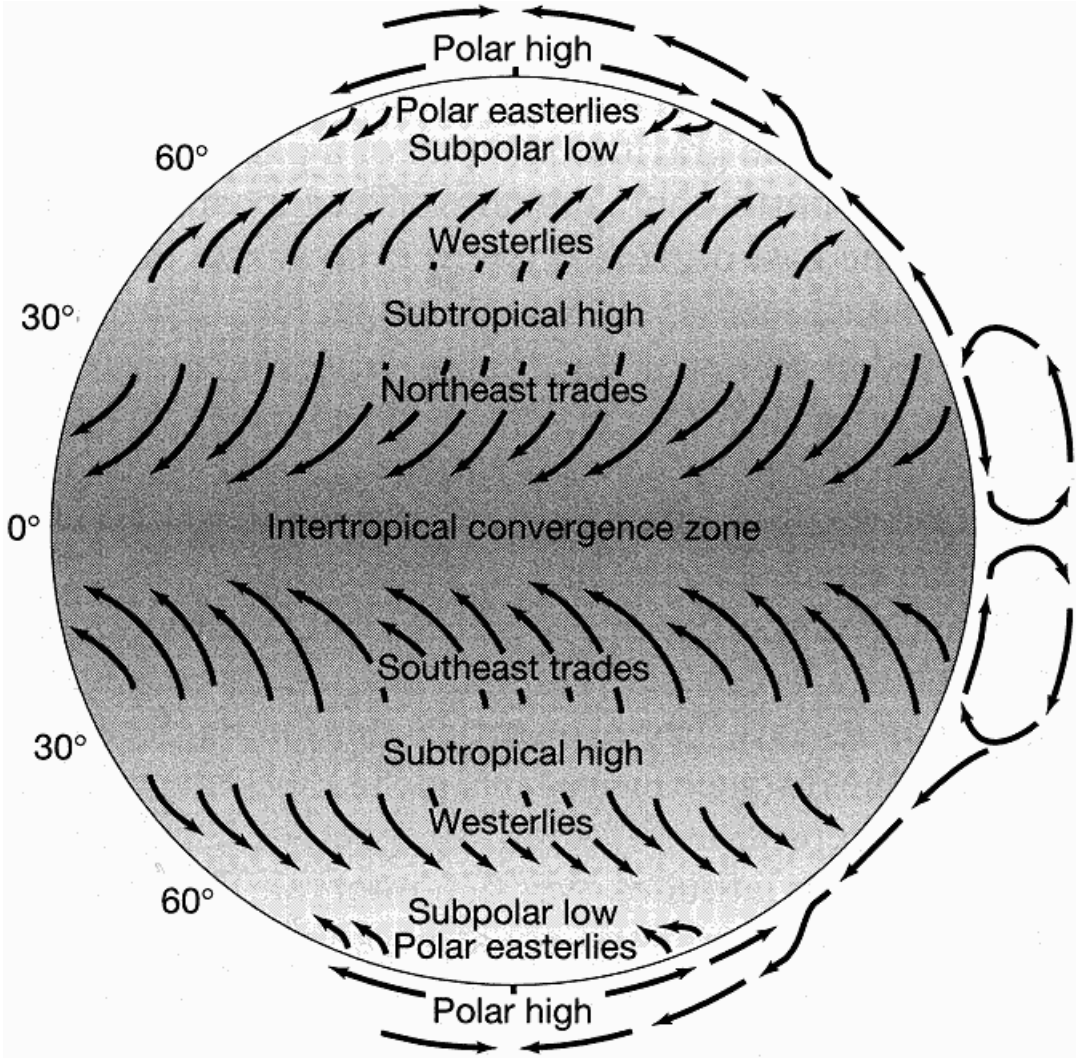
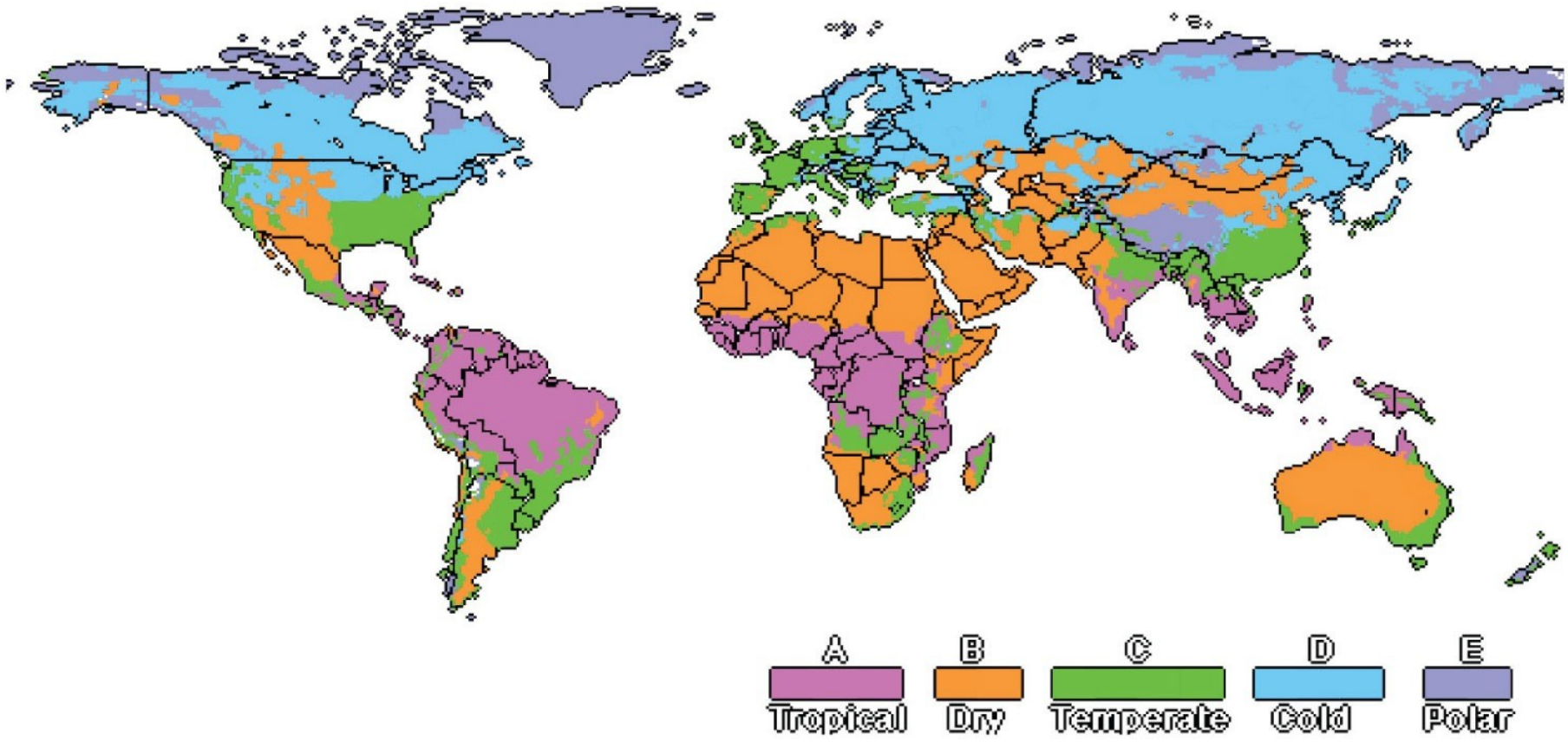
Rising air equals low pressure and the reverse

General Circulation – Surface Winds

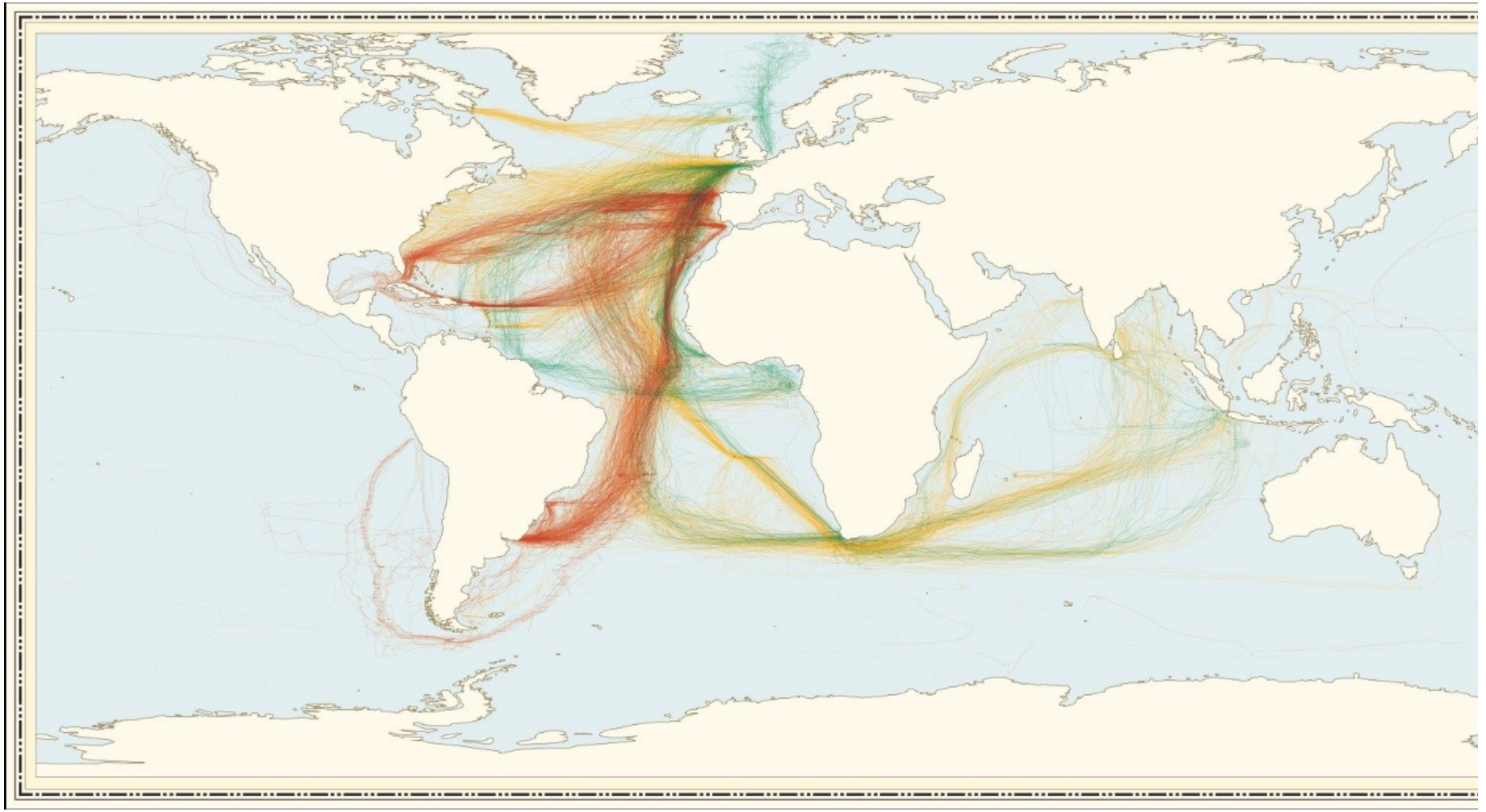


- Surface winds consist of trades and easterlies in the equatorial and polar latitudes, and westerlies at midlatitudes.



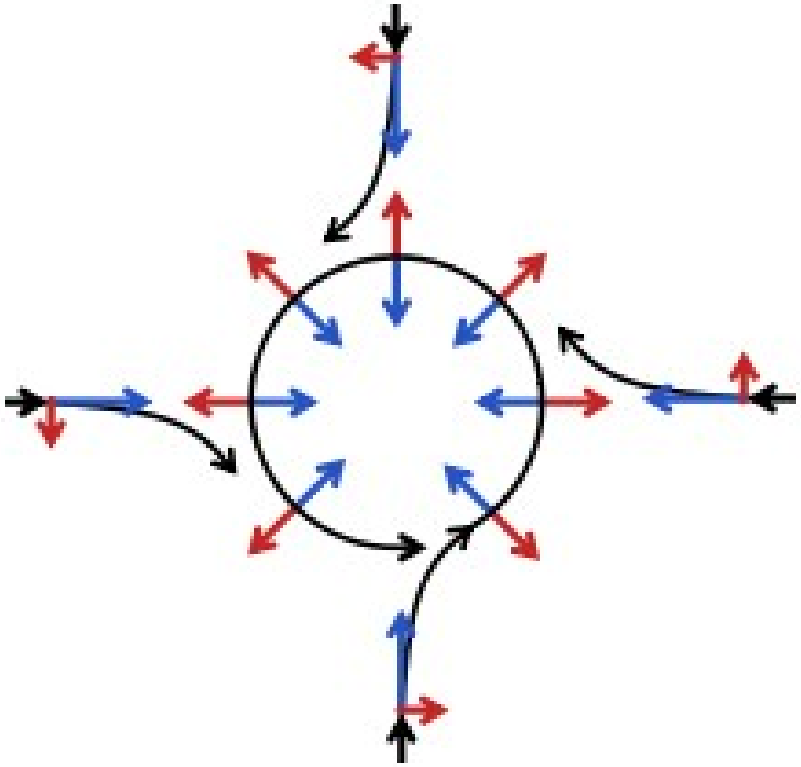
(From Kotteck, M., J. Griieser, C. Beck, B. Rudolf, and R. Rubel. 2006. World map of the Koppen classification updated. *Meteoroligische Zeitschrift* 15: 259–263.)

Shipping Routes 1750-1800



Data source: Climatological Database for the World's Oceans (James Cheshire)This map shows British, Dutch and Spanish shipping routes from 1750 to 1800.

Low Pressure – CounterClockWise (Northern Hemisphere)

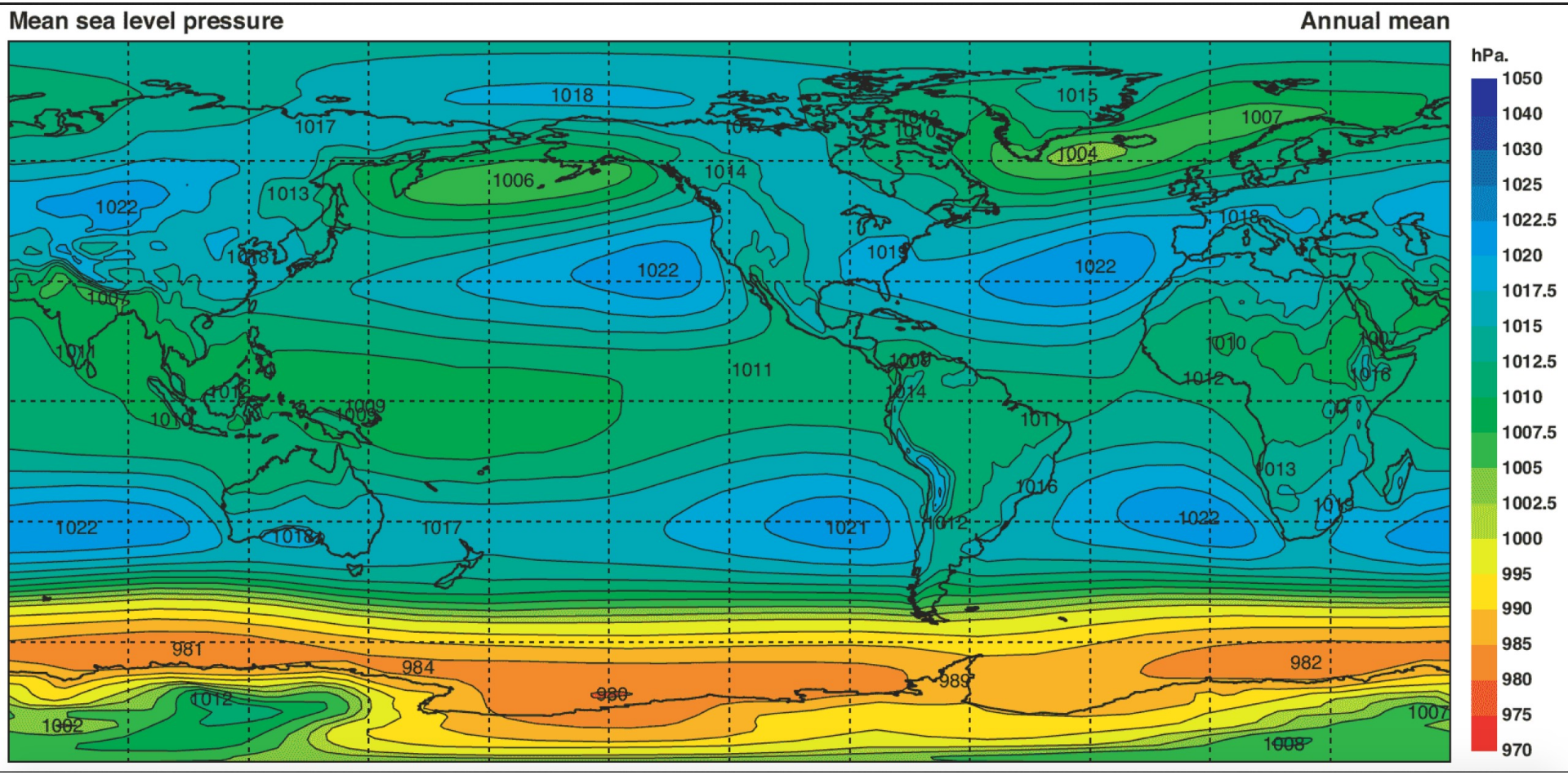


Blue: winds blowing from high to low pressure

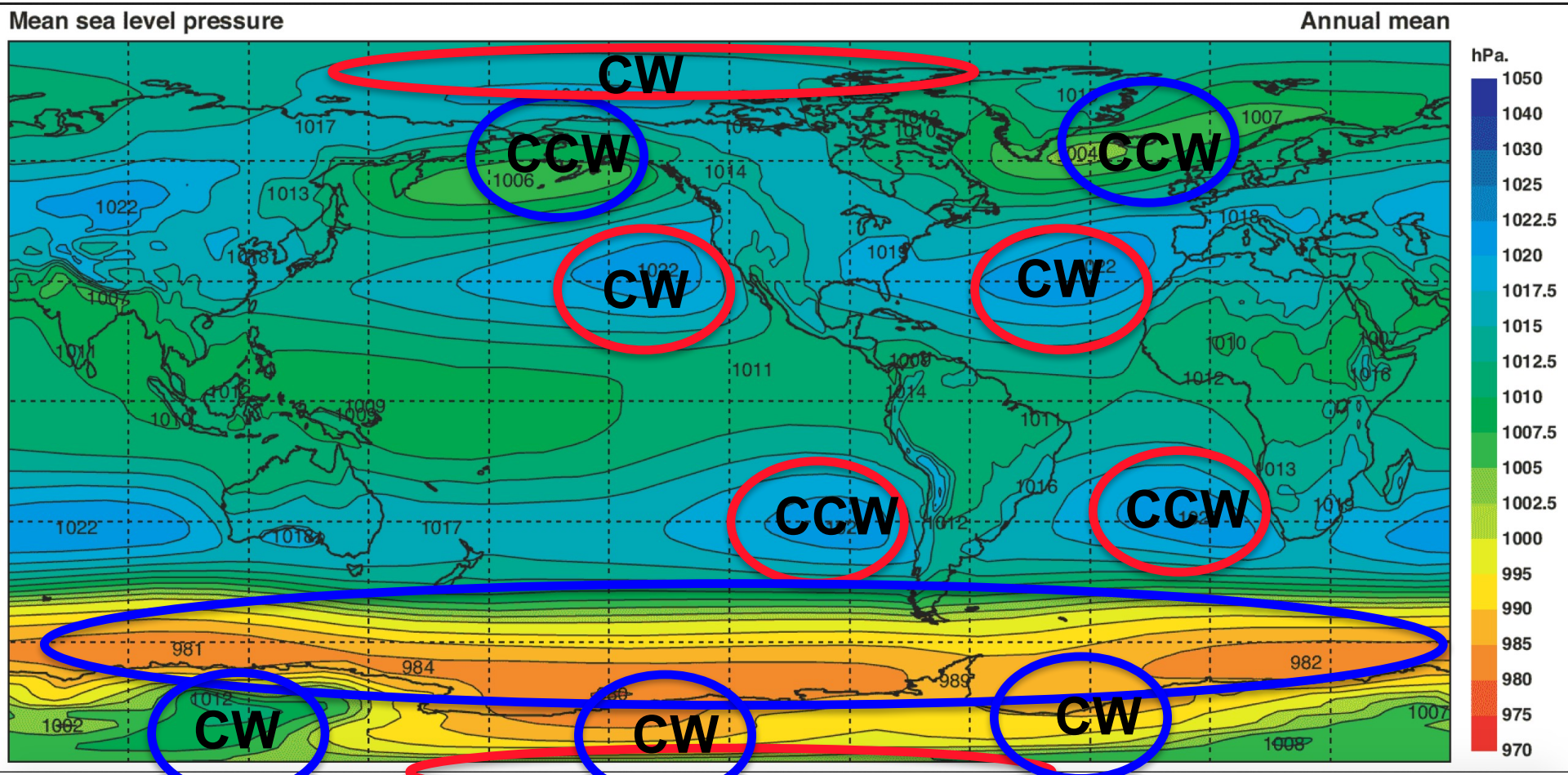
Red: Coriolis effect curving the winds to the right

Black: the path of the winds as the forces balance

Annual Mean Sea Level Pressure



Annual Mean Sea Level Pressure



Jet Streams

- The temperature contrast between the poles and equator drives the westerly winds (**westerlies**) located in the middle latitudes.
- Imbedded within the westerly flow aloft are narrow ribbons of high-speed winds, called **jet streams**, that meander for thousands of kilometers.
- The key to the origin of midlatitude jet streams is found in great temperature contrasts at the surface.
- In the midlatitudes between 30 and 70°, a **polar jet stream** occurs in association with the **polar front**.

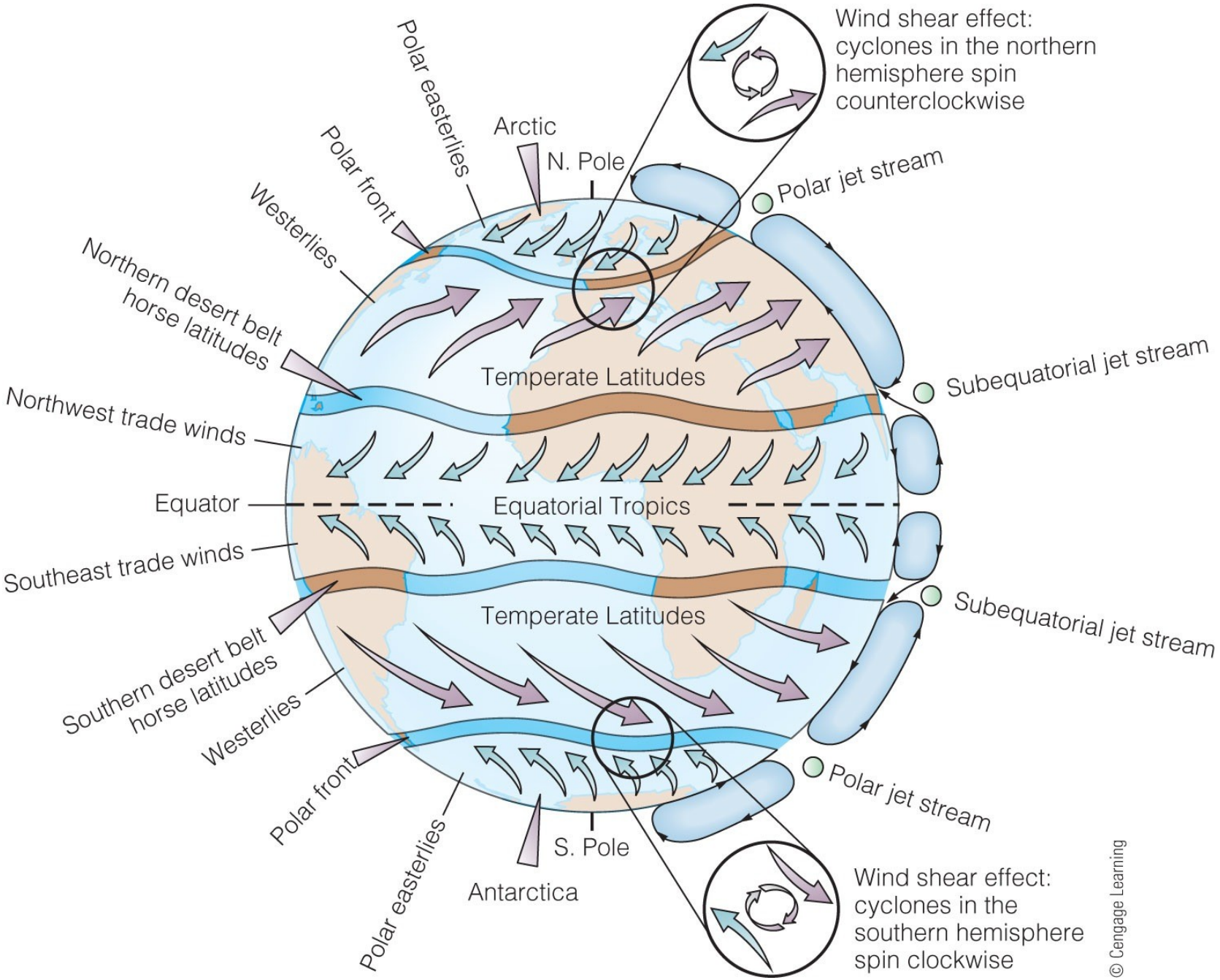
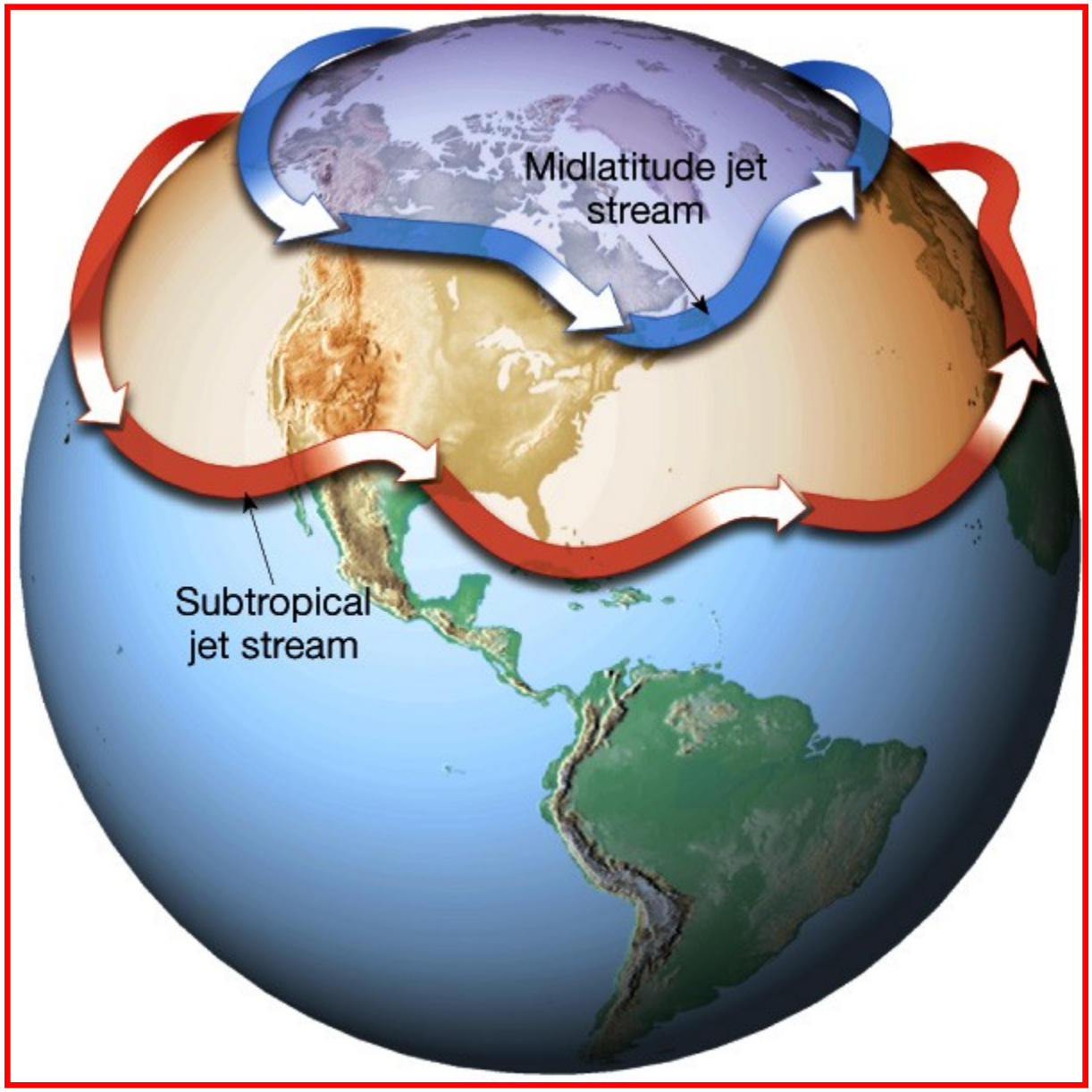
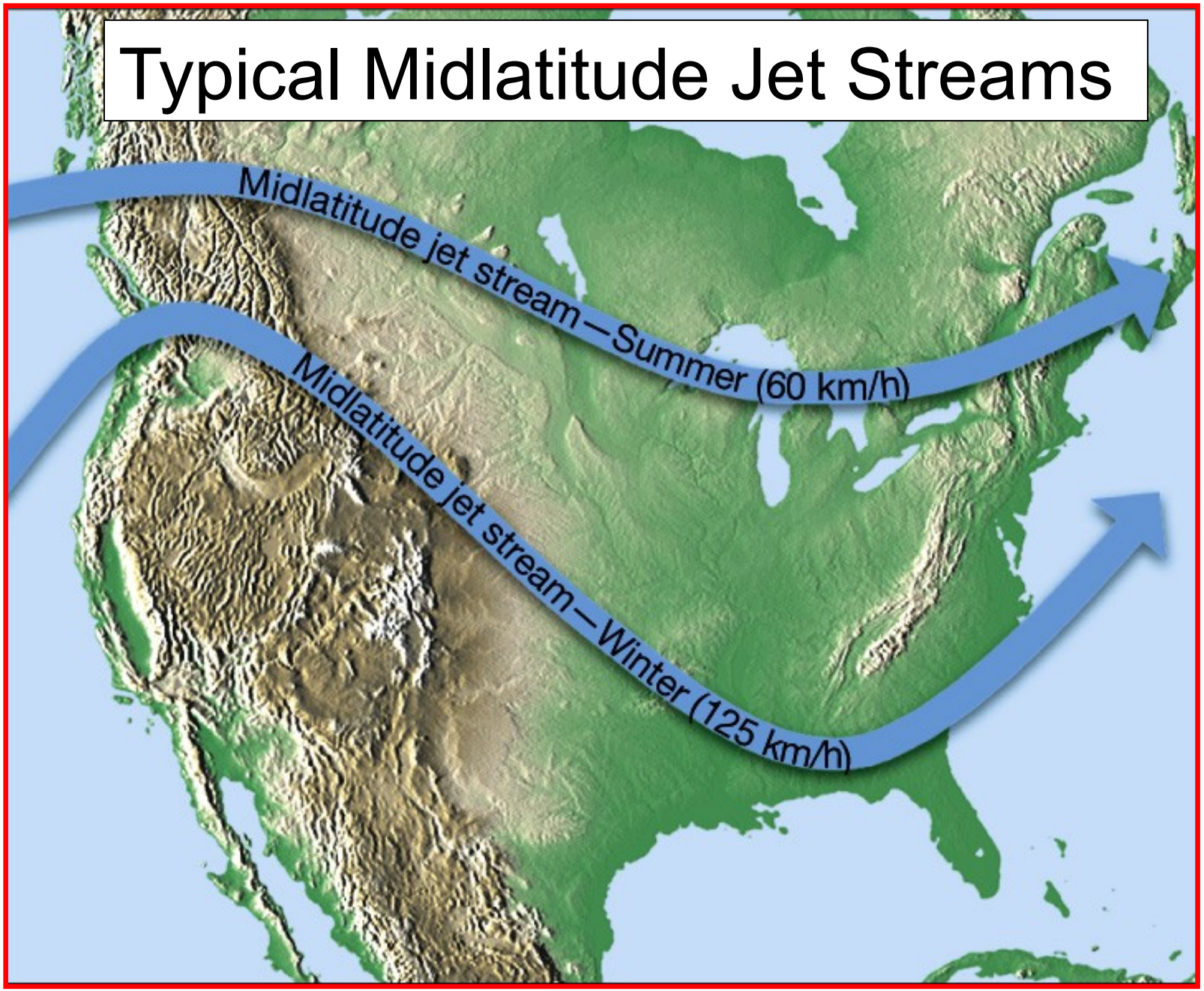
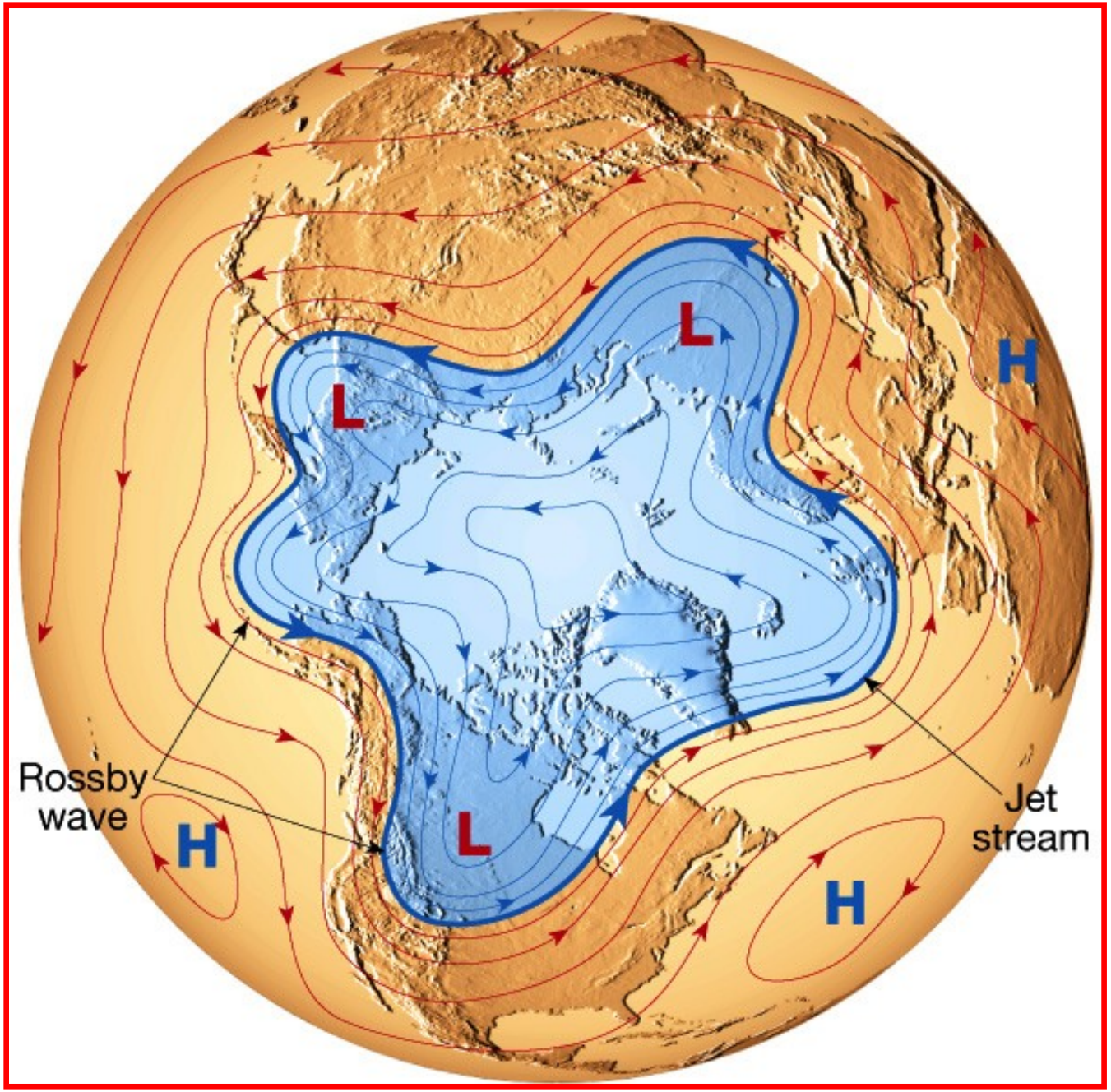


Figure 2-5 p27


Midlatitude jet stream
Subtropical jet stream


Typical Midlatitude Jet Streams
Midlatitude jet stream—Summer (60 km/h)
Midlatitude jet stream—Winter (125 km/h)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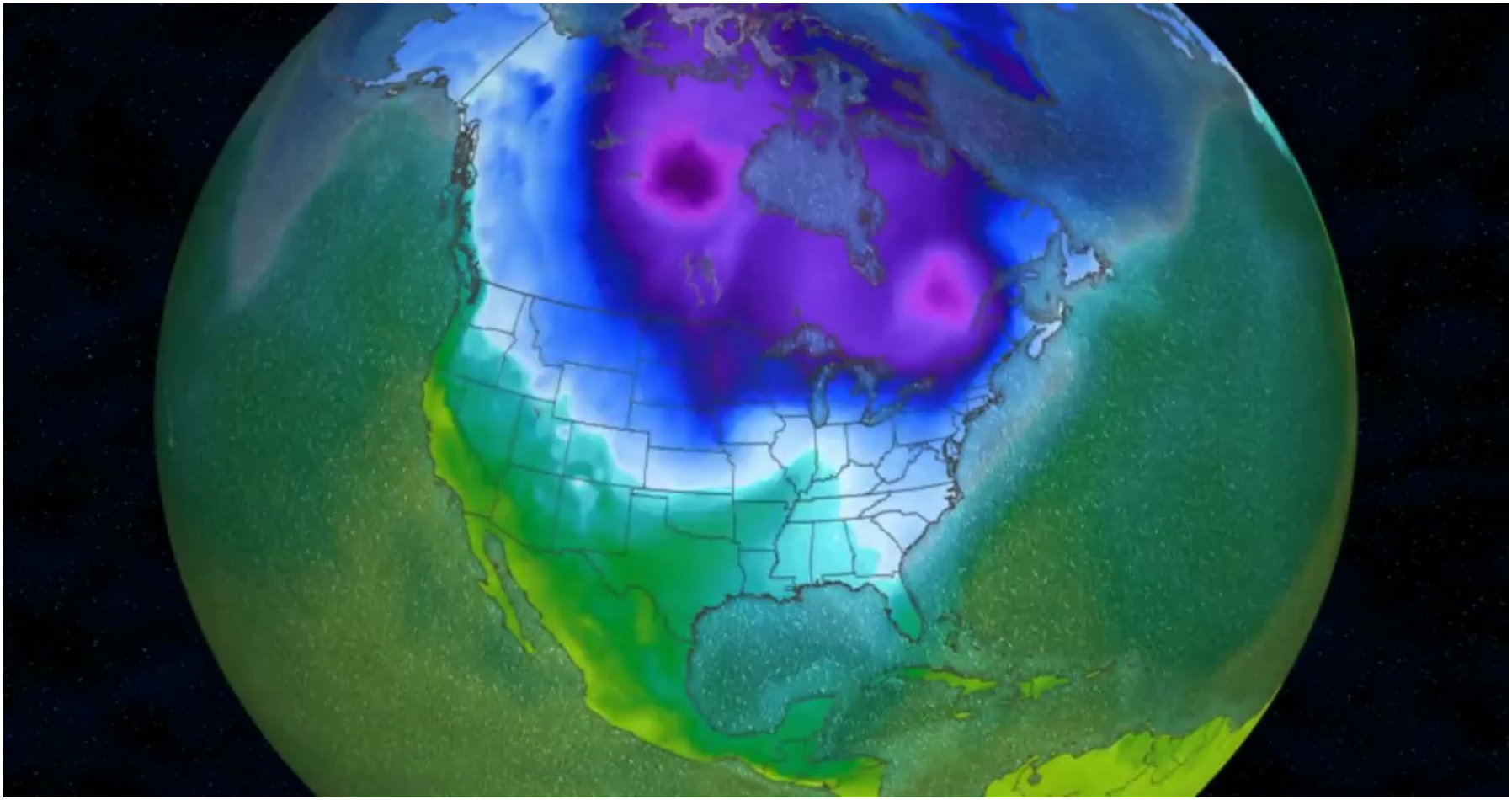
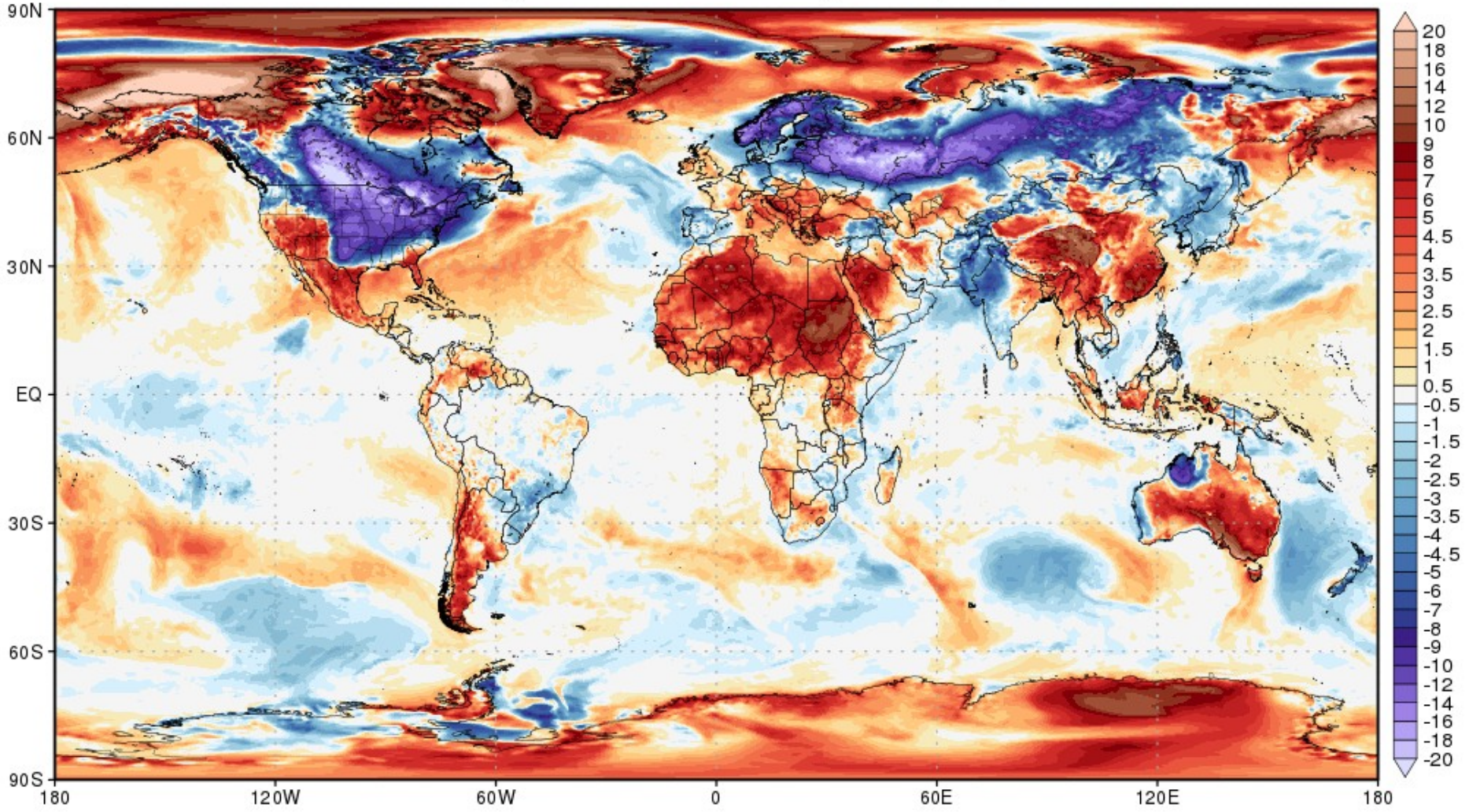
Polar Vortex…



NCEP GFS forecast vs CFSR reanalysis @0.5deg
Run: 21 Jan 2026 12z

Forecast: +66h
Valid: 24 Jan 2026 06z

Temperature anomaly 2m (°C)



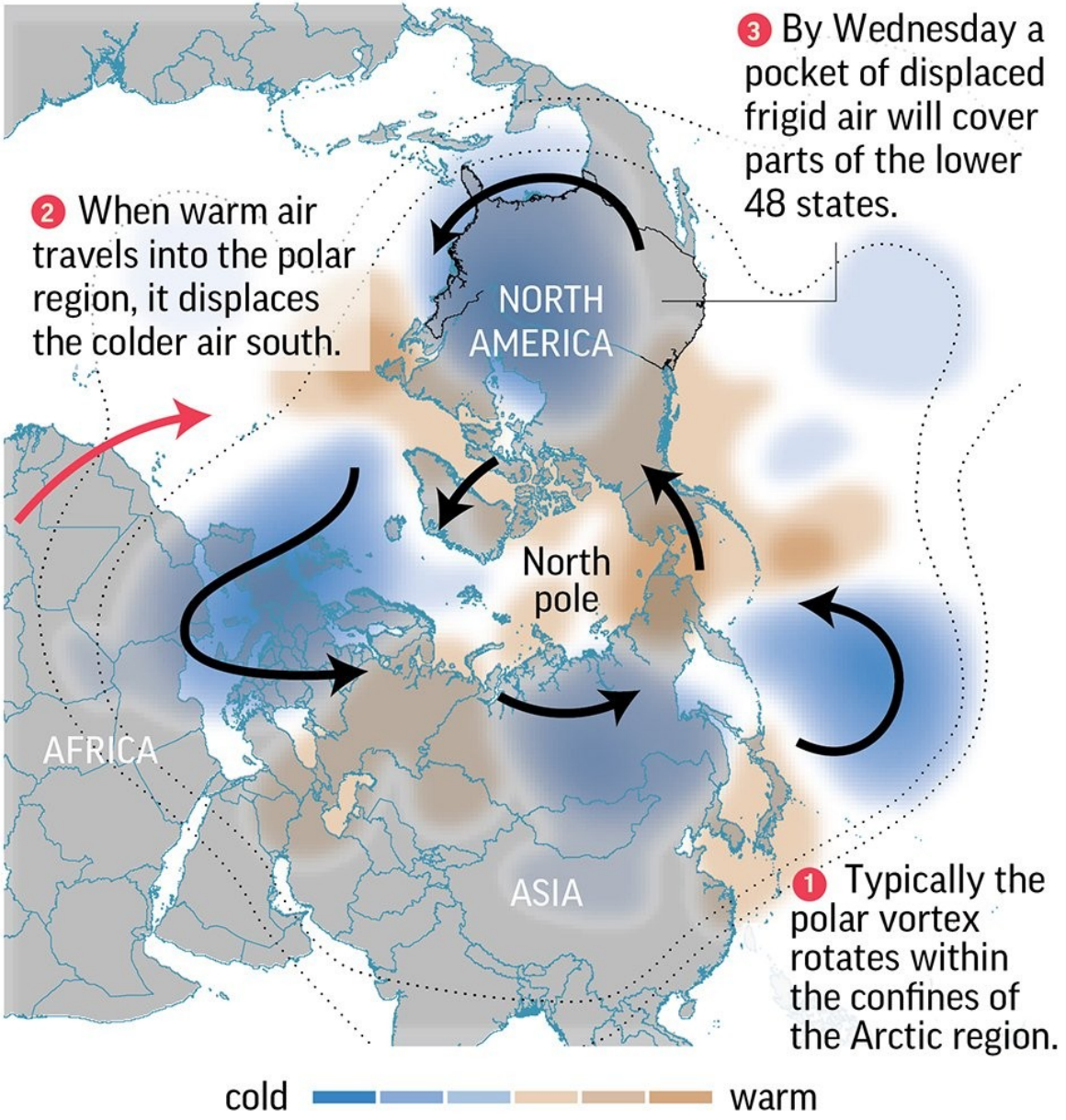
Anomaly global: 0.507K Arctic: 3.044K Tropics: 0.454K Antarctic: 1.340K

(c) Karsten Haustein

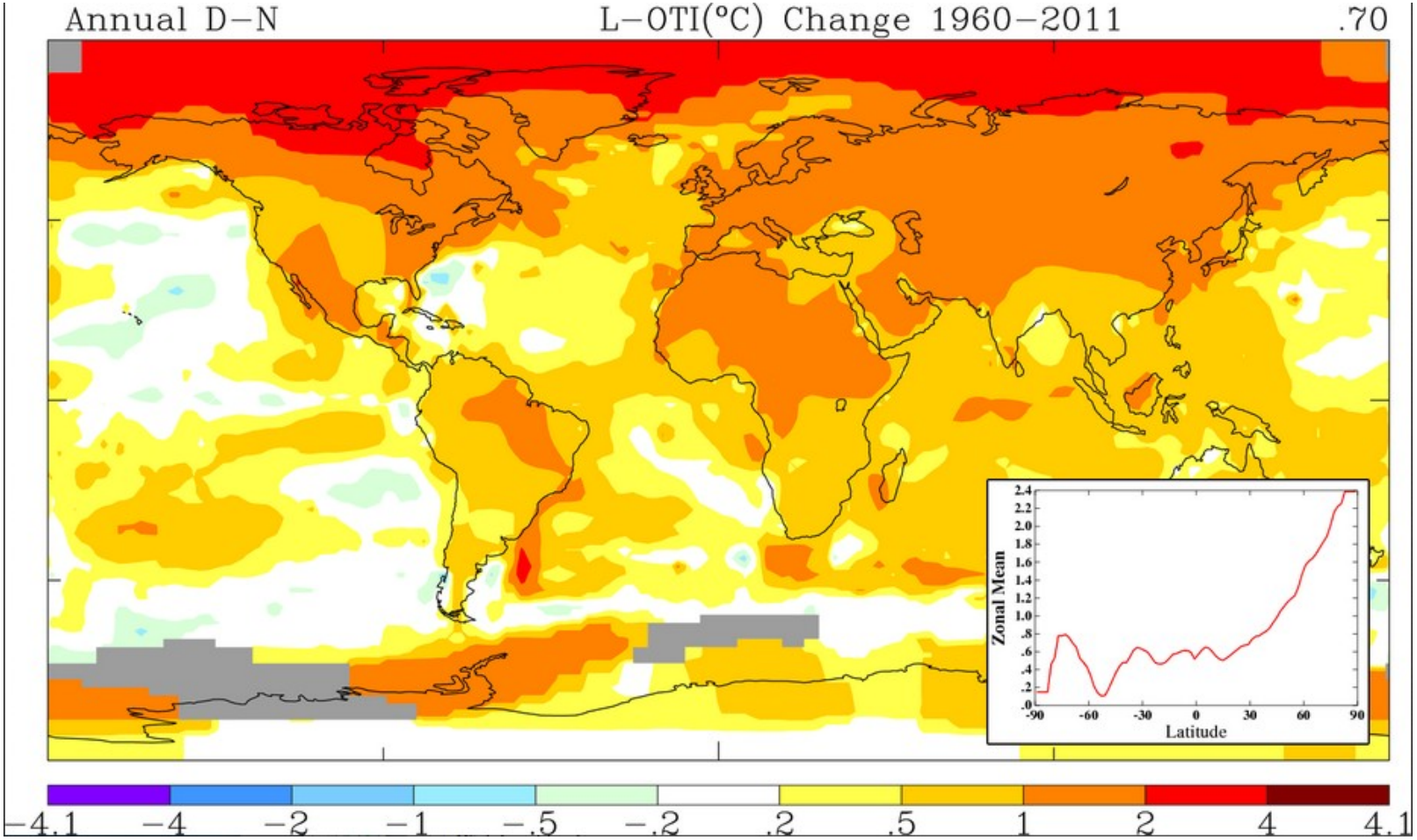
Climatology for 1981-2010 reference period (5 day running mean) | GISS adjusted

Polar Vortex!

Likely will become more common; models suggest that less gradient between equator and pole means a wigglier Jetstream…

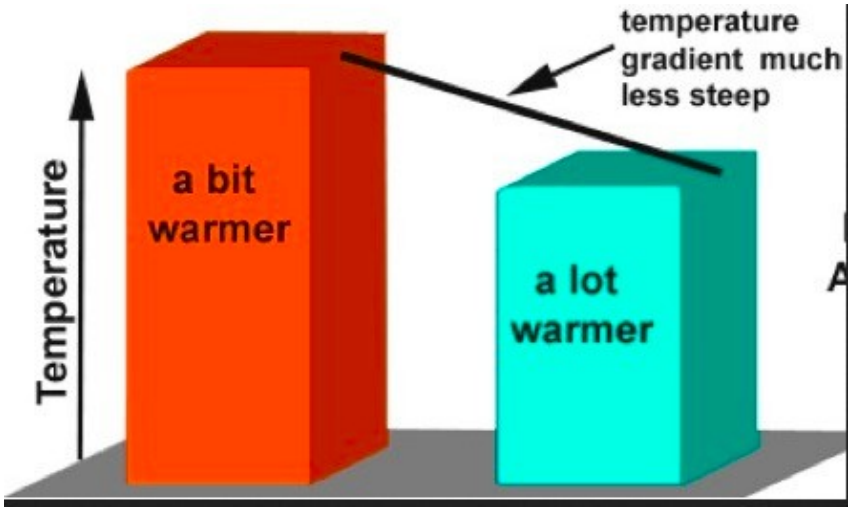


Polar Amplification
Observed

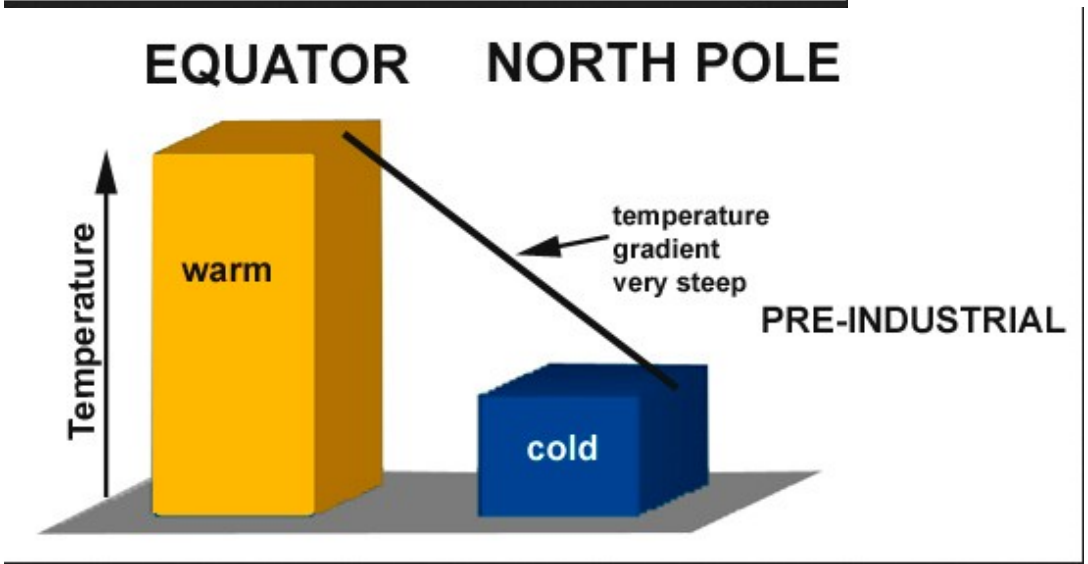


https://nsidc.org/cryosphere/arctic-meteorology/climate_change.html

Equator to Pole Temperature Gradient
Polar Amplification

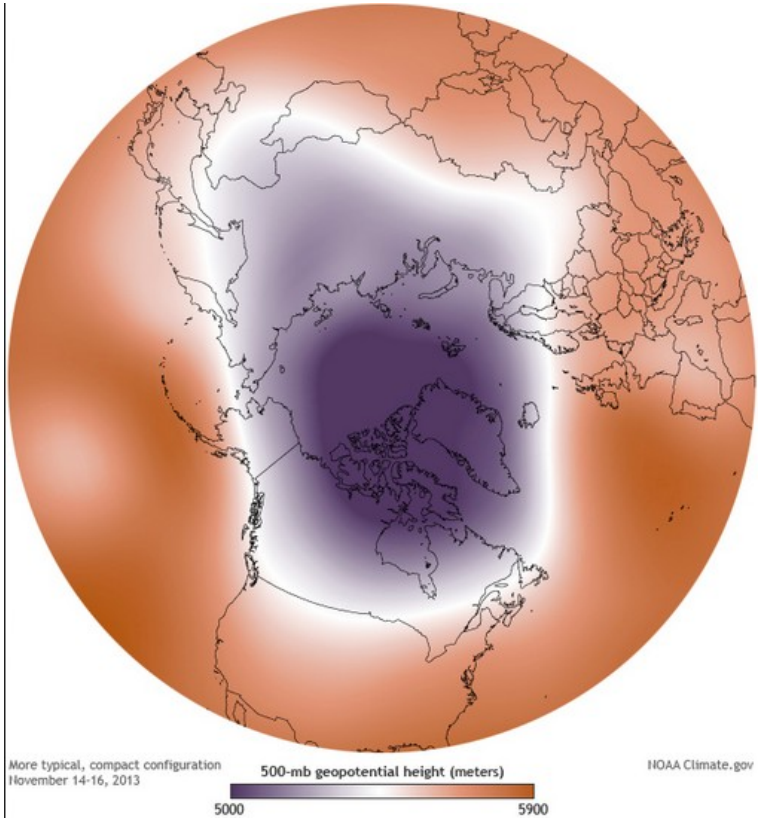


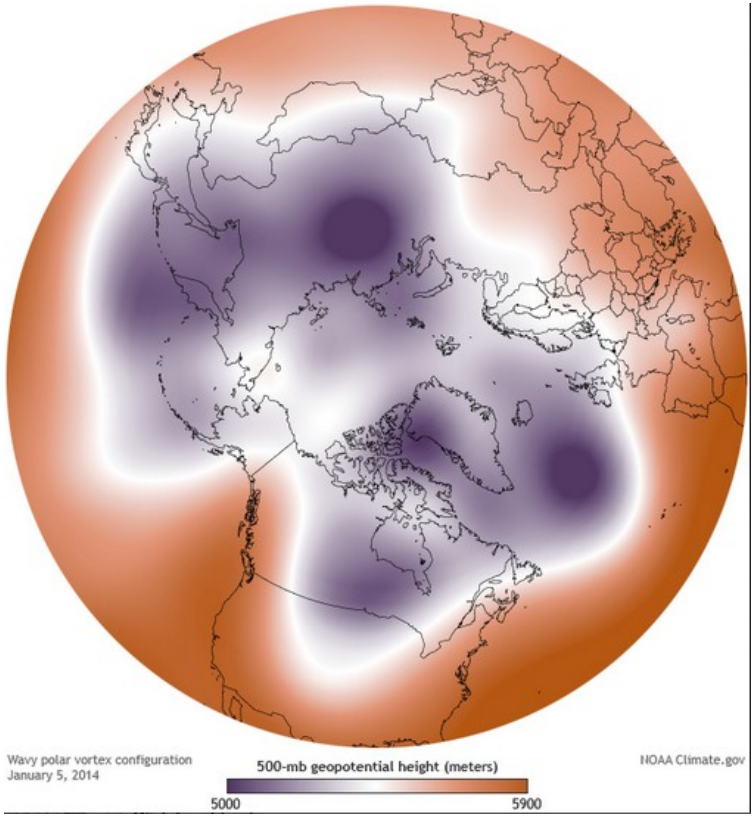
Slower moving
Jet Stream



Fast moving
Jet Stream

Polar Vortex
Cold Core Low

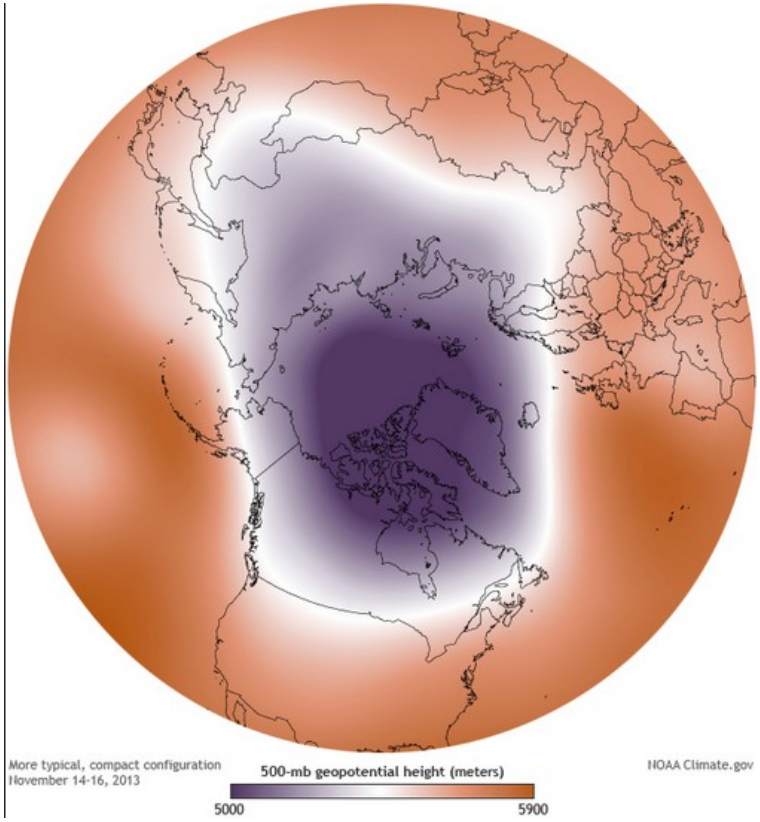


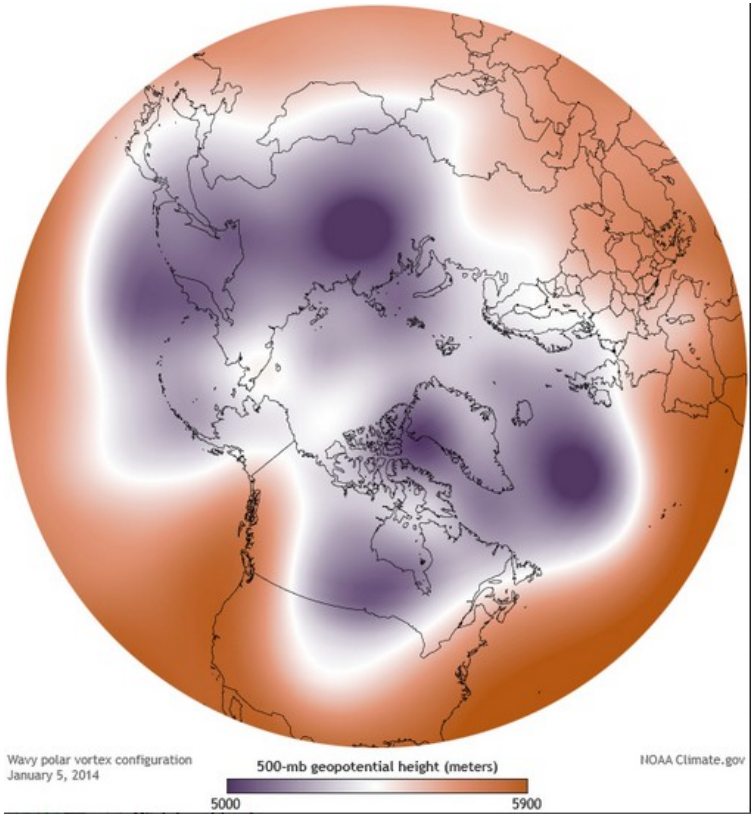


November 2013

January 5, 2014

Polar Vortex
Cold Core Low





Colder Arctic: stronger, more zonal Polar Vortex

→ less cold air outbreaks

Warmer Arctic: weaker, more meandering Polar Vortex

→ more cold air outbreaks